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Paper 1 of 2 — CHE 525

Level 500 • Second Semester • Handout

CHE 525

FUNDAMENTALS OF

RESERVOIR ENGINEERING

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NOMENCLATURE

SYMBOLS DESCRIPTION UNITS

A Area ft^2

B Formation Volume Factor rb/stb

β_o Oil Formation Volume Factor rb/stb

β_g Gas Formation Volume Factor rcf/scf

β_w Water Formation Volume Factor rb/stb

C Isothermal Compressibility $1/\text{psi}$

α_p Pore Compressibility $1/\text{psi}$

α_o Oil Compressibility $1/\text{psi}$

α_w Water Compressibility $1/\text{psi}$

D Vertical (Reservoir) depth ft

e exponential

e^x exponential integral function

E Gas Expansion Factor scf/rcf

ϵ A term in material balance equation which connate H_2O expansion + pore compaction

ϵ_o Expansion + pore compaction = Expansion of oil

ϵ_g term in the material balance equation accounting for the expansion of the gas cap (rb/stb)

ϵ_o term in the material balance equation accounting for the expansion of the oil and its originally dissolved gas (rb/stb)

F underground withdrawal of fluids (rb)

G Gas Initially In Place scf

G_p Cumulative Gas production scf

h Formation thickness m

k Permeability mD

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L Length ft

m Ratio of initial hydrocarbon pore volume of the gas cap to that of the oil column

m Slope of the early linear section of semi-log plot for well test analysis psi/log cycle

m End point mobility ratio number of moles

n Number of moles

N Stock Tank Oil Initially In Place stb

□□ Cumulative Oil Recovery stb

P Pressure Psia

□□ Bubble Point Pressure Psia

□□ Dimensionless Pressure

□□ Pressure @ external boundary Psia

□□ Initial Pressure Psia

□□□ Pressure @ Standard Condition Psia

□□□ Well bore flowing pressure Psia

□□□ Well bore static pressure Psia

□ Average Pressure Psia

q Liquid Production Rate bpd

□□□ Water Injection Rate stb/d

Q Gas Production Rate mmscfd

r Radial distance ft

□□ External Radius ft

□□□ Aquifer/reservoir radius ratio ft

□□ Well bore radius ft

R Production Gas Oil Ratio scf/stb

R Universal gas constant J mol⁻¹ K

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□□ Cumulative Gas Oil Ratio Psia

S Skin factor

□□ Gas saturation

□□□ Residual Gas saturation

□□ Oil saturation

□□□ Residual oil saturation

□□□ Connate Water saturation

t Time hr or yr.

T Temperature OR or K

U Aquifer constant

□ Velocity of flow ft/hr

V Volume ft³

w Width ft

□□ Dimensionless cumulative H₂O

□□ Cumulative water influx

□□ Cumulative water produced

Z Compressibility factor

ABBREVIATION

DST Drill Stem Test

RFT Repeat Formation Tester

BHP Bottom Hole Pressure

FVF Formation Volume Factor

GOR Gas Oil Rate

HCPV Hydrocarbon Pore Volume

PVT Pressure Volume Temperature

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1.0 INTRODUCTION TO RESERVOIR ENGINEERING

1.1 WHAT IS A RESERVOIR?

A reservoir is formed of one (or more) subsurface rock formations containing liquid and/or gaseous hydrocarbons, of usually sedimentary origin. The reservoir rock is porous and permeable, and the structure is bounded by impermeable barriers which traps the hydrocarbons. Figure 1.1 shows a cross section of a typical hydrocarbon reservoir.

Fig 1.1: Cross section of a typical hydrocarbon reservoir

1.2 WHAT IS RESERVOIR ENGINEERING?

- Reservoir engineering may be defined as the application of scientific principles to the drainage problems arising during the development and production of oil and gas reservoirs.
- It has also been defined as “the art of developing and producing oil and gas fluids in such a manner as to obtain a high economic recovery.
- Reservoir Engineering is a part of Petroleum Science that provides the technical basis for the recovery of petroleum fluids from subsurface sedimentary-rock reservoirs.

The Fig. 1.2 below indicates the place and role of Reservoir Engineering in the broad field of Petroleum Science

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Figure 1.2: Reservoir engineering and petroleum science.

1.3 ACTIVITIES IN RESERVOIR ENGINEERING

Reservoir engineering shares the distinction with geology in being one of the great "underground sciences" of the oil industry, attempting to describe what occurs in the wide open spaces of the reservoir between the sparse points of observation - the wells. In applying the subject, it is possible to define four major activities, which are:

- Observations

- Assumptions
- Calculations
- Development decisions

and these can be described as follows.

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1.3.1 Observations:

These include the geological model, the drilling of wells and the data acquired in each: cores, logs, tests, fluid samples. Following the start of field production, the oil, gas and water rates must be continuously and accurately monitored together with any injection of water and gas. Frequent pressure and production logging surveys should also be conducted throughout the lifetime of the project.

It is pointed out that the material balance equation can easily contain eight "unknowns" and failure to collect the essential rate, pressure and pressure-volume-temperature (PVT) data to give oneself a "sporting chance" of attaining a meaningful solution of the equation can make the application of quantitative engineering impossible.

Once the data have been collected and verified, the engineer must interpret them very carefully and collate them from well-to-well throughout the reservoir and adjoining aquifer. This is a most delicate phase of the whole business of understanding reservoirs, in which it can prove dangerous to rely too much on automated techniques. It is fine to use computer programs to interpret well tests, for instance, but to apply sophisticated numerical methods to generalise about formation properties by such means as generating statistical correlations and use of regression analysis should be reduced to a minimum.

1.3.2 Assumptions:

Now the main difficulty arises: having thoroughly examined and collated all the available data, the engineer is usually obliged to make a set of assumptions concerning the physical state of the "system" for which an appropriate mathematical description must be sought.

For instance:

- The oil or gas reservoir is or is not affected by natural water influx from an adjoining aquifer.
- There will or will not be complete pressure equilibrium across the reservoir section under depletion or water drive conditions.

'This is the crucial step in practical engineering. The third activity, calculations, is entirely dependent on the nature of the assumptions made as is the fourth, development decisions, which rely on the results of the calculations.

It is therefore necessary to be extremely cautious when making physical assumptions and the most convincing reservoir studies are those containing the least number. In this respect, the application of material balance is strongly recommended in history matching field performance, as being one of the safest techniques in the business for the simple reason that it requires fewer (far fewer) assumptions than the alternative technique, numerical

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simulation modelling, which should therefore be used as a logical follow-up. Assumptions can usually be confirmed only by observation, not calculation, but this can be problematical.

1.3.3 Calculations:

'While the above comments may seem to denigrate the importance of calculations, that is only meant to be relative to the physical assumptions upon which any mathematical model selected for use is entirely dependent. Once a physical condition has been defined (assumed) then calculations are an absolute must.

1.3.4 Development decisions:

Every action contemplated, planned and executed by reservoir engineers must lead to some form of development decision - otherwise it should not be undertaken in the first place.

1.4 TECHNICAL RESPONSIBILITIES OF THE RESERVOIR ENGINEER

Four distinct technical responsibilities can be defined:

- contributing, with the geologists and petrophysicists, to the estimation of hydrocarbons in place
- determining the fraction of discovered hydrocarbons that can be (reasonably) recovered
- attaching a time scale to the recovery
- day-to-day operational reservoir engineering throughout the lifetime of the project.

1.4.1 Appraisal

The reservoir engineer is involved in the development from the day of discovery of the accumulation and is (or should be) responsible for the collection, collation and interpretation of the following data in the discovery and subsequent appraisal wells.

- Well tests (DSTs)
- Pressure-depth relationships
- Fluid sampling
- Coring

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The most important information gathered from a well test (DST) is the type of hydrocarbon present and whether the formation capable of producing at what may be judged as a commercial rate.

One of the main results determined from pressure-depth surveys, best obtained using the RFT, is the location of fluid contacts (gas-oil, oil-water, gas-water contacts) which are required in the estimation of hydrocarbons in place. The determination of fluid pressure gradients required in such work depends on the collection of reliable fluid samples and their accurate PVT analysis in the laboratory. Also dependent on the PVT analysis is any form of recovery calculation performed and the design of surface facilities to handle separate and dispose of the produced fluids, the oil, gas and water.

Core data is one of the most important requirements from reservoirs in which it is intended that secondary recovery (water-oil or gas-oil) displacement will be the elected means of development. What matters in viewing core data is the all-important permeability distribution across the producing formations; it is this, more than anything else, that dictates the efficiency of displacement processes.

1.4.2 End of appraisal

Usually, the timing of this tends to be gauged by the explorers, when they feel satisfied that they have acquired sufficient data to define within a tolerable range of certainty whether they have proven sufficient stock tank oil initially in place gas initially in place for a viable field development. But this must not be regarded as the sole criterion upon which development decisions are made. The reservoir engineer must be also heavily involved concerning when the appraisal phase should be terminated, based on whether enough meaningful production orientated data have been acquired to perform long-term project development calculations to evaluate:

- Production profiles of oil, gas and water
- Injection profiles of water and/or gas
- Production/injection well]requirements
- Surface topsides facilities (design.

1.4.3 Project Development

This is regarded as the period extending from the time that continuous hydrocarbon production commences until the time of field abandonment. Throughout this time, some of the day-to-day activities of the field reservoir engineer are as follows:

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New well locations, Well completion intervals, Regular well surveys, Areal fluid movement, Production profiles, Decline policy, Enhanced oil recovery

1.5 PRINCIPLES OF RESERVOIR ENGINEERING

The working tools of the reservoir engineer are subsurface geology, applied mathematics, and the basic laws of physics and chemistry governing the behaviour of liquid and vapour phases of crude oil, natural gas, and water in reservoir rock.

To describe 90% of reservoir engineering (excluding thermal effects) requires application of the following physical principles:

- Conservation of mass
- Darcy's law
- Isothermal compressibility
- Newton's second and third laws of motion

1.5.1 Darcy's law

This states that if there is flow through a porous media, there is a relationship for the amount of flow. For one dimensional, linear, horizontal flow (core flooding experiment), Darcy's law (experimental law or empirical relationship) can be expressed in absolute units

as:

$\frac{dp}{dx}$

$= -$

$\frac{q}{k}$

or in field units as

$\frac{dp}{dx}$

$= -1.27 \times 10^{-3} \left(\frac{q}{k} \right)$

$\frac{dp}{dx}$

For linear flow, the pressure gradient $\frac{dp}{dx}$ is intrinsically negative hence the minus sign to make the rate q positive. Note that the state or phase of a fluid in the reservoir usually changes with pressure, the temperature remaining substantially constant.

1.5.2 Isothermal compressibility

Isothermal compressibility accounts for the change in volume of fluids, and the pore volume itself, as the pressure varies at constant temperature. It is defined as:

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$\frac{1}{V} \frac{dV}{dp}$

$= - \left(\frac{dV}{V} \right) / dp$

$\frac{1}{V} \frac{dV}{dp}$

Since the rate of change of volume with respect to pressure is negative, the minus sign is required to assure that the compressibility is positive.

1.5.3 Newton's second and third laws of motion

The second law defines force as equal to mass times acceleration and, in particular, defines the force resulting from the attraction of gravity. Newton's third law of motion states that "action and reaction are equal and opposite". When considering the displacement of one fluid by another with different density, it is imperative that the effect of gravity be accounted for in displacement efficiency calculations in macroscopic reservoir sections. The realism of water displacing oil or gas displacing oil in the field is that the process occurs in hillsides rather than core plugs and in this domain gravity predominates.

1.6 OIL AND GAS RESOURCES AND RESERVES

Reserves are estimated volumes of crude oil, condensate, natural gas, natural gas liquids, and associated substances anticipated to be commercially recoverable from known accumulations from a given date forward, under existing economic conditions, by established operating practices, and under current government regulations. Reserve estimates are based on geologic and/or engineering data available at the time of estimate.

The relative degree of an estimated uncertainty is reflected by the categorisation of reserves as either "proved" or "unproved"

a) Proved Reserves can be estimated with reasonable certainty to be recoverable under current economic conditions. Current economic conditions include prices and costs prevailing at the time of the estimate. Reserves are considered proved if commercial

producibility of the reservoir is supported by actual engineering tests.

b) Unproved Reserves are based on geological and/or engineering data similar to those used in the estimates of proved reserves, but when technical, contractual, economic or regulatory uncertainties preclude such reserves being classified as proved. They may be estimated assuming future economic conditions different from those prevailing at the time of the estimate.

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Figure 1.3: Conceptual scheme for oil and gas resources and reserves.

Unproved reserves may further be classified as probable and possible, see Fig. 1.3.

- Probable Reserves are less certain than proved reserves and can be estimated with a degree of certainty sufficient to indicate they are more likely to be recovered than not.
- Possible Reserves are less certain than proved reserves and can be estimated with a low degree of certainty, insufficient to indicate whether they are more likely to be recovered than not.

The estimation of reserves will depend upon the actual mode of petroleum recovery, which may involve either a natural-drive mechanism improved by water or gas injection, or some special technique of enhanced oil recovery (EOR).

1.7 PETROLEUM RESERVES ESTIMATION METHODS

The process of estimating oil and gas reserves for a producing field continues throughout the life of the field. There is always uncertainty in making such estimates. The level of uncertainty is affected by the following factors:

1. Reservoir type,

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2. Source of reservoir energy,
3. Quantity and quality of the geological, engineering, and geophysical data,
4. Assumptions adopted when making the estimate,
5. Available technology, and
6. Experience and knowledge of the evaluator.

Figure 1.4: Magnitude of uncertainty in reserves estimates

The magnitude of uncertainty, however, decreases with time until the economic limit is reached and the ultimate recovery is realized, see Figure 1.4.

The oil and gas reserves estimation methods can be grouped into the following categories:

1. Analogy,
2. Volumetric,
3. Decline analysis,
4. Material balance calculations for oil reservoirs,
5. Material balance calculations for gas reservoirs,
6. Reservoir simulation.

1.7.1 Analogy-Based Approach

In the early stages of development, reserves estimates are restricted to the analogy and volumetric calculations. Another producing reservoir with comparable characteristics can be used as a possible analogue for the reservoir under consideration, either by a direct well-to-well comparison or on a unit recovery basis. This can be done by determining an average oil or gas recovery per well in the analogue reservoir (e.g., 100,000 bbl/well) and applying a similar or adjusted recovery factor to the wells in the reservoir considered.

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In an analogue approach, one has to consider similarities of well spacing, reservoir rock lithofacies, rock and fluid properties, reservoir depth, pressure, temperature, pay thickness and drive mechanism. All possible differences between the analogue reservoir and the reservoir in question need to be considered to make a realistic adjustment of the recovery estimates. However, an analogue-based approach is the least accurate and little reliable method of petroleum reserve estimation, simply because perfect analogues can seldom be found.

1.7.2 Volumetric Estimates

The volumetric method, on the other hand, entails determining the areal extent of the reservoir, the rock pore volume, and the fluid content within the pore volume. This provides an estimate of the amount of hydrocarbons-in-place. The ultimate recovery, then, can be estimated by using an appropriate recovery factor. Each of the factors used in the calculation above have inherent uncertainties that, when combined, cause significant uncertainties in the reserves estimate.

The methods of reserve estimation based on reservoir data are volumetric and can be divided into deterministic and probabilistic (stochastic) estimates. The main difficulty in a volumetric estimate of resources/reserves is in the transfer of data obtained at a small scale (core analysis, lithofacies data, well logs, etc.) into a much larger scale (i.e. data "upscaling" for interwell space).

A comparison of the deterministic and probabilistic methods, however, can provide quality assurance for estimating hydrocarbon reserves; i.e. reserves are calculated both deterministically and probabilistically and the two values are compared. If the two values agree, then confidence on the calculated reserves is increased. If the two values are away different, the assumptions need to be re-examined.

1.7.2.1 Deterministic Methods

The deterministic method is by far the most common. The procedure is to select a single value for each parameter to input into an appropriate equation, to obtain a single answer.

The principle of a deterministic approach to resources/reserve estimates is to "upscale" the information derived from the wells and supported by seismic survey, into the interwell space by using an interpolation technique.

The main parameters used for a volumetric estimate in this approach are:

- The reservoir "gross" isopach map, which means the bulk thickness of the reservoir rocks (formation).

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- The reservoir "net" isopach map, which means the cumulative thickness of the permeable rock units only. The Net-to-Gross ratio (N/G) is an important parameter indicating the productive portion of the reservoir.
- The reservoir rock porosity
- The permeability and net-thickness product ($kxhN$) is important for the estimation of well production capacity
- Volume-based average saturation of water, gas and oil

Plotting these parameters as contoured maps (isopachs, isoporosity, isopermeability, etc.) provides the crucial information on their variation and distribution in the reservoir and makes it possible to evaluate the reservoir pore volume and its fractions saturated with oil and gas (hydrocarbon volume).

1.7.2.2 Stochastic Methods

The probabilistic method is more rigorous and less commonly used. This method utilizes a distribution curve for each parameter and, through the use of Monte Carlo Simulation; a distribution curve for the answer can be developed. Assuming good data, a lot of qualifying information can be derived from the resulting statistical calculations, such as the minimum and maximum values, the mean (average value), the median (middle value), the mode (most likely value), the standard deviation and the percentiles,

The probabilistic estimation of resources/reserves takes more account of the estimate uncertainty. Stochastic reservoir description is usually based on the procedure of random-number generator. This numerical technique assumes that the main reservoir properties (porosity, permeability, N/G, etc.) all have random, possibly normal, frequency distributions, with the range of values included by core and well-log data. The maximum and minimum values are specified for each of the reservoir parameters and the random number generator then "drowns data", so to speak, and then simulates their actual density distribution in the whole reservoir.

In practice, it is necessary to repeat the stochastic simulation for different "seeds" (initial boundary values) in order to assess and quantify the actual variation of a given parameter. Each numerical realisation bears an uncertainty for the reservoir characterisation, where the probabilistic rather than deterministic, is an estimate of resources/reserves.

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1.7.3 Performance Analysis

The methods of performance analysis presently used include:

- Analysis based on Material Balance Equation (MBE)
- Reservoir Simulation Models (RSM)
- Decline Curve Analysis (DCA)

The aim of all of these methods is to obtain the best reservoir performance prediction on the basis of available data. As production and pressure data from a field become available, decline analysis and material balance calculations, become the predominant methods of calculating reserves. These methods greatly reduce the uncertainty in reserves estimates; however, during early depletion, caution should be exercised in using them.

The MBE method is based on the data obtained from previous reservoir performance and PVT analysis, but involves some assumptions for the reservoir driving mechanism in order to minimise the range of possible predictions from the dataset. The method is thus adjusted differently to reservoirs containing oil, gas or oil with a gas cap (primary or secondary).

The RSM method involves a numerical simulation technique, with the matching of the production and the reservoir's previous performance (history). The discrepancy between the simulation results (prediction) and the available data is minimised by adjusting the reservoir parameters and taking into account the most likely reservoir drive mechanism (history match).

The DCA method is to predict future performance of the reservoir by matching the observed trend of the production decline with one or several standard mathematical methods of the production rate-time (hyperbolic, harmonic, exponential, etc.). If successful, such a performance analysis allows to estimate both the reserves and the future performance of the reservoir.

Decline curve relationships are empirical, and rely on uniform, lengthy production periods. It is more suited to oil wells, which are usually produced against fixed bottom-hole pressures. In gas wells, however, wellhead back-pressures usually fluctuate, causing varying production trends and therefore, not as reliable.

The following "decline curves" from production wells are commonly used in the DCA:

- Production rate vs. time.
- Production rate vs. cumulative oil production.
- Water cut vs. cumulative oil production.
- Gas-oil ratio vs. cumulative production.
- Percentage oil production vs. cumulative oil production.
- The (p/z) ratio vs. cumulative gas production.

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Some of these decline curves are shown in Fig. 1.5.

Figure 1.5: Different ways of data representation for a decline curve analysis.

1.8 PRODUCTION PHASES

The first phase, production build up, which may exist or not depending on the drilling strategy is the increased production as wells are brought on stream. Clearly, as in some cases, wells might be predrilled through a template and then all brought on stream together when connected to production facilities, such a build up of production will, therefore, not occur.

The next stage represents the period when the productivity of the production facility is at its design capacity and the wells are throttled back to limit their productivity. This period is called the plateau phase when production is maintained at the design capacity of the facilities. Typical production rates for the plateau period cannot be presented since it depends on the techno-economics of the field.

A time will come when the reservoir is no longer able to deliver fluids to match the facilities capacity and the field goes into the decline phase. This phase can be delayed by methods to increase production. Such methods could include artificial lift, where the effort required to

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lift the fluids from the reservoir is carried out by a downhole pump or by using gas lift to reduce the density of the fluid system in the well.

There comes a time when the productivity of the reservoir is no longer able to generate revenues to cover the costs of running the field, This abandonment time again is influenced by the size and nature of the operation.

Figure 1.6: Phases in production.

1.9 OIL RECOVERY TECHNIQUES

1.9.1 Primary Oil Recovery

This is the recovery obtained through the natural energy of the reservoir. The Natural Drive Mechanism types include:

1. Depletion Drive: A depletion type reservoir is a reservoir in which the hydrocarbons contained are NOT in contact with a large body of permeable water bearing sand. In a depletion type reservoir the reservoir is virtually totally enclosed by porous media and the only energy comes from the reservoir system itself.

In depletion drive reservoirs the energy comes from the expansion of the fluids in the reservoir and its associated pore space. There are two types of depletion drive reservoirs, solution gas drive reservoirs and gas cap drive reservoirs.

- Solution Gas Drive: In solution gas drive reservoirs the initial condition is where the reservoir is undersaturated, i.e. above the bubble point. Production of fluids down to the bubble point is as a result of the effective compressibility of the system. This combined compressibility (of oil, water and pores) provides the drive mechanism for depletion drive above the bubble point. Solution gas drive only occurs once the bubble point pressure has been reached.

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- Gas Cap Drive: Another kind of depletion type is where there is already free gas in the reservoir, accumulated at the top of the reservoir in the form of a gas cap. This gas cap drive reservoir, as it is termed, receives its energy from the high compressibility of the gas cap. The two significant sources of driving energy are gas cap expansion and expansion of gas coming out of solution.

2. Water Drive: reservoir is one in which the hydrocarbons are in contact with a large

volume of water bearing sand.

3. Gravity drainage: or gravitational segregation is a situation where the natural density segregation of the phases is responsible for moving the fluids to the well bore.

Gravity drainage is where the relative density forces associated with the fluids cause the fluids, the oil, to drain down towards the production well.

4. Compaction Drive: Although not a common drive energy, the characteristics of its occurrence can be dramatic. Compaction drive occurs when the hydrocarbon formation is compacted as a result of the increase in the net overburden stress as the reservoir pore pressure is reduced during production.

1.9.2 Secondary Recovery

This is considered when the energy is supplemented by injection of fluids, for example gas or water, to maintain the pressure or partially maintain the pressure. The injected fluid also acts as a displacing fluid sweeping the oil to the producing wells.

1.9.3 Tertiary or Enhanced Oil Recovery (EOR)

After sweeping the reservoir with water or gas there will still be remaining oil; oil at a high saturation where the water for a range of reasons, for example; well spacing, viscosity, reservoir characteristics to name just a few, has by-passed the oil. The oil which has been contacted by the injected fluid will not be completely displaced from the porous media. Because of characteristics of the rock and the fluids a residual saturation of fluid is held within the rock. Both of these unrecovered amounts, the by-passed oil and the residual oil are a target for enhanced recovery methods, EOR. Figure 1.7 gives a schematic representation of the various phases of development and includes the various improved recovery methods

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Figure 1.7 Oil Recovery Mechanisms.

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2.0 FUNDAMENTALS OF RESERVOIR ROCKS AND FLUID PROPERTIES

2.1 ROCK PROPERTIES

Rock properties are determined by performing laboratory analyses on cores from the reservoir to be evaluated. There are basically two main categories of core analysis tests that are performed on core samples regarding physical properties of reservoir rocks. These are Routine core analysis tests and Special tests:

Routine core analysis tests

- Porosity
- Permeability
- Saturation

Special tests

- Overburden pressure
- Capillary pressure

- Relative permeability
- Wettability
- Surface and interfacial tension

The above rock property data are essential for reservoir engineering calculations as they directly affect both the quantity and the distribution of hydrocarbons and, when combined with fluid properties, control the flow of the existing phases (i.e., gas, oil, and water) within the reservoir.

Petrophysics is the study of the physical properties of rocks. For a rock to be a reservoir rock, it must fulfill the following characteristics:

- It must have a certain storage capacity: this property is characterized by the porosity
- The fluids must be able to flow in the rock: this property is characterized by the permeability
- It must contain a sufficient quantity of hydrocarbons, with a sufficient concentration: the impregnated volume is a factor here, as well as the saturations.

The methods used to characterize reservoir rocks are essentially core analysis and well logging. Rock properties are determined by performing laboratory analyses on cores from the reservoir to be evaluated

The specifications of a reservoir rock are such that there must be a large enough capacity to store economically viable amounts of hydrocarbon and the hydrocarbon must flow at economical rates when penetrated by a well. The factors which may affect the capacity and the flow properties are the porosity, permeability, capillary pressure, compressibility and fluid saturation.

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In order to contain enough oil or gas to make production economically viable, a reservoir rock must exceed: a minimum porosity, a minimum thickness, a minimum permeability, and a minimum area.

2.1.1 Porosity

Porosity is one of the most important rock properties as a measure of the space available for accumulation of hydrocarbon fluids. The porosity of a porous medium is given the symbol of ϕ and is defined as the ratio of void space, or pore volume (pore volume), to the total bulk volume (bulk volume) of the rock. This definition refers to an absolute porosity. This ratio is expressed either as a fraction or in percent.

Equation

$\phi =$

Equation

The factors which affect the porosity (and permeability) of sandstone include the packing, size and shape of the grains, variations in size of grains, arrangement in which grains were laid down and compacted, and amount of clay and other materials which cement the sand grains together.

2.1.1.1 Absolute and effective porosity

Absolute porosity

The absolute porosity is defined as the ratio of the total pore space in the rock to that of the bulk volume. A rock may have considerable absolute porosity and yet have no conductivity to fluid for lack of pore interconnection.

Effective porosity

The effective porosity is the percentage of interconnected pore space with respect to the bulk volume. The effective porosity is the value that is used in all reservoir engineering calculations because it represents the interconnected pore space that contains the recoverable hydrocarbon fluids.

2.1.2 Permeability

The permeability of a rock is the description of the ease with which fluid can pass through the pore structure. In order to extract the fluids the rock must be permeable which requires that there be sufficiently large, interconnecting pores.

In engineering the unit of permeability is the Darcy, derived from the empirical equation known as Darcy's Law named after a French scientist who investigated the flow of water through filter beds in 1856. His work provided the basis of the study of fluid flow through porous rock.

Darcy's law of fluid flow states that rate of flow through a given rock varies directly with the pressure applied, the area open to flow and varies inversely with the viscosity of the fluid

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flowing and the length of the porous rock. In terms of equating the parameters, the constant of proportionality in the equation is termed the permeability.

$$Q \propto \frac{\Delta p}{L}$$

$$Q =$$

$$\frac{\Delta p}{L}$$

where:

Q = flow rate in cm³/sec

A = cross sectional area of sample in cm²

Δp = pressure different across sample, atm

μ = viscosity in centipoise

L = length of sample in cm

k = permeability in Darcy

2.1.3 Saturation

Saturation is defined as that fraction, or percent, of the pore volume occupied by a particular fluid (oil, gas, or water). This property is expressed mathematically by the following relationship:

$$S_o = \frac{V_o}{V_p} \quad S_g = \frac{V_g}{V_p} \quad S_w = \frac{V_w}{V_p}$$

$$\frac{V_{\text{res}}}{V_{\text{res}}} = \frac{V_{\text{res}}}{V_{\text{res}}}$$

Applying the above mathematical concept of saturation to each reservoir fluid gives

$$\frac{V_{\text{res}}}{V_{\text{res}}} = \frac{V_{\text{res}}}{V_{\text{res}}}$$

$$\frac{V_{\text{res}}}{V_{\text{res}}} = \frac{V_{\text{res}}}{V_{\text{res}}}$$

$$\frac{V_{\text{res}}}{V_{\text{res}}} = \frac{V_{\text{res}}}{V_{\text{res}}}$$

The saturation of each individual phase ranges between zero to 100%. By definition, the sum of the saturations is 100%, therefore

$$S_{\text{g}} + S_{\text{o}} + S_{\text{w}} = 1$$

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2.2 THERMODYNAMIC PROPERTIES OF PURE HYDROCARBON COMPOUNDS AND MIXTURES

Hydrocarbon mixtures may exist in either the gaseous or liquid state, depending on the reservoir and operating conditions to which they are subjected. To understand and predict the volumetric behaviour of oil and gas reservoirs as a function of pressure, knowledge of the physical properties of reservoir fluids must be gained.

The most important of these properties are:

- Critical pressure, p_c
- Critical temperature, T_c
- Critical volume, V_c
- Critical compressibility factor, z_c
- Acentric factor, ω
- Molecular weight, M

These fluid properties are usually determined by laboratory experiments performed on samples of actual reservoir fluids. In the absence of experimentally measured properties, it is necessary for the petroleum engineer to determine the properties from empirically derived correlations.

(a) Katz and Firoozabadi (1978) presented a generalized set of tabulated physical properties, such as average boiling point, specific gravity, and molecular weight etc, for the petroleum fractions C6 through C45.

(b) Ahmed (1985), then correlated Katz-Firoozabadi-tabulated physical properties with the number of carbon atoms of the fraction by using a regression model. The generalized equation has the following form:

$$\rho = a_1 + a_2 n + a_3 n^2 + a_4 n^3 + (a_5/n) \quad (1-1)$$

where ρ = any physical property

n = number of carbon atoms, i.e., 6, 7, . . . , 45

$a_1 - a_5$ = coefficients of the equation and are given in Table 1-3

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(c) Riazi and Daubert (1987) developed a simple two-parameter equation for predicting the physical properties of pure compounds and undefined hydrocarbon mixtures. The proposed generalized empirical equation is based on the use of the molecular weight M and specific gravity of the undefined petroleum fraction as the correlating parameters. Their mathematical expression has the following form:

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(d) Edmister (1958) proposed a correlation for estimating the acentric factor of pure fluids and petroleum fractions. The equation, widely used in the petroleum industry, requires boiling point, critical temperature, and critical pressure. The proposed expression is given by the following relationship:

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Exercise

- 1.
- 2.

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(1) CORRELATIONS FOR PHYSICAL PROPERTIES OF RESERVOIR FLUIDS

There are well-established physical property correlations for the following reservoir fluids:

- Natural gases
- Crude oil systems
- Reservoir water systems

Natural gases

Generally, the natural gas is a mixture of hydrocarbon and non-hydrocarbon gases. The hydrocarbon gases that are normally found in a natural gas are methanes, ethanes, propanes, butanes, pentanes, and small amounts of hexanes and heavier. The non-hydrocarbon gases (i.e., impurities) include carbon dioxide, hydrogen sulfide, and nitrogen.

Knowledge of pressure-volume-temperature (PVT) relationships and other physical and chemical properties of gases is essential for solving problems in natural gas reservoir engineering. These properties include:

- Apparent molecular weight, M_a
- Specific gravity, ρ_g

- Compressibility factor, z
- Density, ρ_g
- Specific volume, v
- Isothermal gas compressibility coefficient, c_g
- Gas formation volume factor, B_g
- Gas expansion factor, E_g
- Viscosity, μ_g

The above gas properties may be obtained from direct laboratory measurements or by prediction from generalized mathematical expressions.

(i) If we assume an ideal situation, then the PVT relationships are given by:

Ideal gas law equation

$$pV = nRT$$

where p = absolute pressure, psia

V = volume, ft³

T = absolute temperature, °R

n = number of moles of gas, lb-mole

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R = the universal gas constant which, for the above units, has the

value 10.730 psia ft³/lb-mole °R

The density of the gas can be given by:

where ρ_g = density of the gas, lb/ft³

(ii) Assuming non-ideal situation; since natural gas is not a pure component but a mixture of hydrocarbon components, the overall physical and chemical properties can be determined from the physical properties of the individual components in the mixture by using appropriate mixing rules.

The basic properties of gases are commonly expressed in terms of the apparent molecular weight, standard volume, density, specific volume, and specific gravity.

These properties are defined as follows:

Apparent Molecular Weight

Standard Volume

The standard volume is then defined as the volume of gas occupied by 1 lb-mol of gas at standard conditions of 14.7 psia and 60°F.

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Density

Specific Volume

The specific volume is defined as the volume occupied by a unit mass of the gas.

Specific Gravity

The specific gravity is defined as the ratio of the gas density to that of the air; with both densities measured at the standard pressure P_{sc} and standard temperature T_{sc}

or

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Example

1. Three pounds of n-butane are placed in a vessel at 120°F and 60 psia. Calculate the volume and density of the gas assuming ideal gas behaviour. Given that the molecular weight of n-butane is, $M = 58.123$

Solution

2. A gas well is producing gas with a specific gravity of 0.65 at a rate of 1.1MMscf/day. The average reservoir pressure and temperature are 1,500 psi and 150°F. Calculate:

- Apparent molecular weight of the gas
- Gas density at reservoir conditions
- Flow rate in lb/day

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Solution

-
-
-

3. A gas well is producing a natural gas with the following composition:

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Assuming ideal gas behaviour, calculate:

- Apparent molecular weight
- Specific gravity
- Gas density at 2000 psia and 150°F
- Specific volume at 2000 psia and 150°F

Solution

- Applying
-
-
-

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Real Gas Situation

At high pressure and high temperature, the PVT relationship is more accurately represented by:

where the gas compressibility factor z is a dimensionless quantity and is defined as the ratio of the actual volume of n -moles of gas at T and p to the ideal volume of the same number of moles at the same T and p :

Studies of the gas compressibility factors for natural gases of various compositions have shown that compressibility factors can be generalized with sufficient accuracies for most engineering purposes when they are expressed in terms of the following two dimensionless properties:

- Pseudo-reduced pressure
- Pseudo-reduced temperature

$$Z = f(P_{pr}, T_{pr})$$

These dimensionless terms are defined by the following expressions:

where

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P = system pressure, psia

P_{pr} = pseudo-reduced pressure, dimensionless

T = system temperature, °R

T_{pr} = pseudo-reduced temperature, dimensionless

P_{pc} , T_{pc} = pseudo-critical pressure and temperature, respectively, and defined by the following relationships:

Standing and Katz (1942) presented a generalized gas compressibility factor chart as shown in Figure 2-1. The chart represents compressibility factors of sweet natural gas as a function of P_{pr} and T_{pr} . This chart is generally reliable for natural gas with minor amount of non-hydrocarbons. It is one of the most widely accepted correlations in the oil and gas industry.

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Example 1

A gas reservoir has the following gas composition: the initial reservoir pressure and temperature are 3000psia and 180°F, respectively.

Calculate the gas compressibility factor under initial reservoir conditions.

Solution

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Determine the z-factor from Figure 2-1, to give:

$$z = 0.85$$

Example 2

Using the data below and assuming real gas behaviour, calculate the density of the gas phase under initial reservoir conditions.

Compare the results with that of ideal gas behaviour.

Solution

From the chart

Given that

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Calculate the density of the gas assuming an ideal gas behaviour ($Z = 1$)

The results of the above example show that the ideal gas equation estimated the gas density with an absolute error of 15% when compared with the density value as predicted with the real gas equation.

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PSEUDO-CRITICAL PRESSURE AND PSEUDO-CRITICAL TEMPERATURE OF GASES WHERE COMPOSITION OF A NATURAL GAS IS NOT AVAILABLE

Standing (1977) provided a mathematical expression of the graphical correlation by Brown et al. (1948) for a convenient approximation of the pseudo-critical pressure and pseudo-critical temperature of gases when only the specific gravity of the gas is available.

Example 3

Using the data below and assuming unavailability of pseudo-critical temperature and pseudo-critical pressure data, calculate the density of the gas phase under initial reservoir conditions.

Given that for a natural gas system

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Solution

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Absolute and Effective Porosity

Absolute porosity

Effective porosity

Application of the effective porosity

One important application of the effective porosity is its use in determining the original hydrocarbon volume in place. Consider a reservoir with an areal extent of A acres and an average thickness of h feet. The total bulk volume of the reservoir can be determined from the following expressions:

The reservoir pore volume PV can then be determined by combining Equations 4-4 and 4-5 with 4-3.

NB: 1 Acre = 43,560 square feet and 1 barrel = 5.614583 cubic feet

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Example 1

An oil reservoir exists at its bubble-point pressure of 3000 psia and temperature of 160°F. The oil has an API gravity of 42° and gas-oil ratio of 600 scf/STB. The specific gravity of the solution gas is 0.65. The following additional data are also available:

- Reservoir area = 640 acres
- Average thickness = 10 ft
- Connate water saturation = 0.25
- Effective porosity = 15%

Calculate the

(a) Oil Initially In Place (OIIP) in bbl

(b) Stock Tank Oil Initially In Place (STOIIP) in STB.

Solution

(a) Determine the specific gravity of the stock-tank oil from

Calculate the initial oil formation volume factor (FVF) by applying Standing's equation,

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Calculate the pore volume from

Pore volume = $7758 (640) (10) (0.15) = 7,447,680$ bbl

OIIP = Pore Volume (1 - Swc)

OIIP = 7,447,680 (1 - 0.25) = 5,585,760 bbl

(b) Calculate the Stock Tank Oil Initially In Place (STOIIP).

STOIIP = Pore Volume (1 - Swc)

Boi

Initial oil in place = 7,447,680 (1 - 0.25)/1.306 = 4,276,998 STB

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2.2 COMPOSITIONAL DESCRIPTION FOR RESERVOIR ENGINEERING

Petroleum engineers require a compositional description tool to use as a basis for predicting reservoir and well fluid behaviour. The two approaches that are commonly used are the multicomponent compositional model and the two component black oil model. The latter simplistic approach has been used for many years to describe the composition and behaviour of reservoir fluids. It is called the "Black Oil Model".

2.2.1 Black Oil Model

Two component description of the reservoir fluid consisting of gas dissolved in oil (solution gas) and stock tank oil. Compositional changes with varying pressure and temperature are ignored. To those appreciating thermodynamics this simplistic two component model is difficult to cope with.

Figure 2.1: "Black Oil Model"

Terms such as 'Gas Oil Ratio' and 'Formation Volume Factor' are black oil model terms. Associated with the black oil model are Black Oil model definitions in relation to Gas Solubility and Formation Volume Factors.

2.2.2 Compositional Model

The compositional model is based on the paraffin series C_nH_{2n+2} . To keep the number of components in the model manageable, long chain members are grouped together and given an average property. These compounds are termed collectively as the 'C+ fraction'. Typically

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this covers the hydrocarbons above Heptane and therefore is called the C7+ fraction, which is characterised using the terms Apparent Molecular Weight and Specific Gravity.

Figure 2.2: Compositional Model

2.2.3 Comparison of Reservoir Fluid Models

It is useful to summarise the differences between the Black Oil Model approach compared to the Compositional Model in predicted fluid properties.

The suitability of the two approaches is largely related to the nature of the fluid. For heavier oils where there are low GOR's as compared to volatile oils with high GOR's, black oil models are likely to be suitable. For the more volatile systems where there are more significant compositional variations in a reservoir as pressure is depleted, compositional models are considered more capable of predicting fluid behaviour. The computational requirements of

compositional models used to be a restriction when carrying out large reservoir simulation. The continued development of computing and associated equations of state modelling reduces these former restrictions.

The list below gives a summary comparison of the two approaches.

Black Oil Models

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- Components - solution gas and stock tank oil
- Bo, Rg, etc.
- Empirical correlations

Compositional Models

- N components based on paraffin series
- Equation of state based calculations

2.3 PVT FLUID PROPERTIES

Inspection and rationalisation of the pressure-volume-temperature (PVT) fluid properties must be the opening move in the study of any oilfield since the PVT functions, which relate surface to reservoir volumes, are required in practically every aspect of reservoir engineering: calculation of hydrocarbons in place, pressure depth regimes, any form of recovery calculations and to assure correct design of surface facilities.

2.3.1 BASIC PVT PARAMETERS

For an oil reservoir these are:

- BO: oil formation volume factor (FVF), which is the number of reservoir barrels of oil and dissolved gas that must be produced to obtain one stock tank barrel of stable oil at the surface (rb/stb).
- RS: solution or dissolved gas/oil ratio (GOR), which is the number of standard cubic feet of gas produced with each stock tank barrel of oil that was dissolved in the oil in the reservoir (scf/stb).
- Bg: gas formation volume factor, which is the volume in barrels that one standard cubic foot of gas at the surface occupies as free gas in the reservoir (rb/scf).

All three are strictly functions of pressure at the (assumed) constant reservoir temperature.

The first two allow for the fact that oil in the reservoir, at high pressure and temperature, contains dissolved gas that is released as it is produced to the surface. For example, if BO = 1.45 rb/stb and RS = 500 scf/stb, then 1.45 rb of oil and dissolved gas must be produced to obtain 1 stb of oil, releasing 500 scf of gas at the surface. As mentioned above, all three functions are required to relate reservoir volumes to those measured at the surface. The engineer works physically in terms of reservoir volumes yet must use practical surface measured volumes in equations. Thus, in setting-up the basic equation of waterdrive in terms of: injection rate = gross production rate (at constant pressure), it becomes:

$$Q_{inj} = Q_{oil} + Q_{gas} \left(\frac{B_g}{B_o} \right)$$

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in which all the rates, q , are surface measured values (stb/d) but multiplication by the FVFs (r_b /stb), establishes the equation as an "underground" balance (r_b /d), in which q_o is the flow rate of the single phase oil containing R , (scf/stb) of dissolved gas while q_{wR} is the reservoir flow rate of the produced water which may contain a small amount of dissolved gas, hence the need for the water FVF B_W , which is usually close to unity.

All three PVT parameters are measured in laboratory experiments conducted at the constant reservoir temperature. The results are then dependent on the declining pressure alone.

Fig. 2.3. PVT functions for a moderate volatility oil. (a) FVF; (b) solution GOR.

Typical shapes of the BO and RS functions for a moderate volatility oil are shown in Fig. 2.3. At the initial pressure, p_i , the oil is described as "undersaturated", meaning that if there was free gas available it would be dissolved - but there is none. The oil is therefore in the single phase state containing dissolved gas. Eventually, through continued depletion, the pressure drops to the bubble point, p_b , at which the oil becomes gas saturated. Below this pressure gas is liberated from the oil giving rise to the presence of two phases: oil + dissolved gas and free gas.

Above the bubble point, as the pressure declines, the oil must expand hence BO increases. Since the oil compressibility is usually fairly small, however, the extent of the expansion is slight. Also, above the bubble point each stb of oil produced contains the constant R_{si} scf of dissolved gas, since none has been liberated in the reservoir.

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Figure 2.4: Production of reservoir hydrocarbons (a) above bubble point pressure, (b) below bubble point pressure

Below the bubble point, matters are more complicated. Gas is freed from the oil in situ, consequently B_o decreases continuously since the dissolved gas content is reduced. The decrease in oil volume from its maximum value at the bubble point is referred to as the "shrinkage", $B_{ob} - B_o$, at which point the amount of gas released is $R_{si} - R_{so}$ scf/stb. Producing below the bubble point is invariably "messy", because the gas viscosity is typically fifty times lower than that of oil, therefore, according to Darcy's law, it is capable of travelling much faster. This leads to recovery under the so-called "solution gas drive" condition. It would be wrong to state that production below the bubble point necessarily results in a low oil recovery but on most occasions it does. There is an uncontrollable element associated with the process due to the completely different mobilities of the oil and free gas, which leads to the tendency to strip the gas from the reservoir prematurely at the expense of the less mobile oil

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Figure 2.5: Application of PVT parameters to relate surface to reservoir hydrocarbon volumes; above bubble point pressure.

Figure 2.6: Application of PVT parameters to relate surface to reservoir hydrocarbon volumes; below bubble point pressure

In volatile oil reservoirs, in which the oil contains a large volume of dissolved gas that will be released at the surface (typically: $BO_i = 2.8$ rb/stb, $RS_i = 3000$ scf/stb) the variation in FVF with pressure is as shown in Fig. 2.7. Even above the bubble point, the high dissolved gas content

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in the oil means that the change in volume with pressure is very significant. This is reflected in the oil compressibility which might typically have a value of $60-80 \times 10^{-6}$ /psi. Problems start, however, immediately below the bubble point where the shrinkage, $BO - BO_i$, is very dramatic and, as the oil shrinks in situ, it gives the gas greater freedom to move through the formation and be produced in excess quantities. The situation can be so extreme that what was accepted as an oil field one day can appear to behave like a gas field the next, simply by passing through the bubble point pressure. This is reflected in the highly unfavourable oil relative permeability which inhibits its flow. Therefore, in such volatile oil reservoirs it is obviously preferable to aim at operating in the undersaturated pressure range just above the bubble point (to take advantage of the natural oil expansion) through the injection of either water or gas to provide pressure maintenance.

Fig. 2.7. FVF of a volatile oil.

The third PVT parameter, the gas FVF, B_g (rb/scf), is only required in initially undersaturated reservoirs once gas has been liberated below the bubble point pressure. It then relates free gas volumes in the reservoir (rb) to the surface volume (scf). Alternatively, if there is a gas cap at initial conditions, B_g would be required from the start. On account of its units (rb/scf) the FVF has a very small value at high pressure, typically, 0.0007 rb/scf. While this should not be of concern to us these days, as a historical note, it is worth mentioning the problems it posed our predecessors in the subject: armed computationally with only slide rules, with which the difficulty was trying to figure out where the decimal point should be located! The B_g function has a rather unfortunate, near hyperbolic shape (Fig. 2.8), causing some difficulties in interpolation/extrapolation. When working in the low-pressure range, the engineer should be careful to choose very small pressure increments when entering the B_g function to computer programs to assure accuracy in its use.

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Fig. 2.8. Gas FVF as a function of pressure

2.4. COLLECTION OF FLUID SAMPLES (SAMPLING RESERVOIR FLUIDS)

Samples of the reservoir fluid are usually collected at an early stage in the reservoir's producing life and dispatched to a laboratory for the full PVT analysis. There are basically two ways of collecting such samples, either by direct subsurface sampling or by surface recombination of the oil and gas phases. Whichever technique is used the same basic problem exists, and that is, to ensure that the proportion of gas to oil in the composite sample is the same as that existing in the reservoir. Thus, sampling a reservoir under initial conditions, each stock tank barrel of oil in the sample should be combined with RS_i standard cubic feet of gas i.e. 1 stb Oil + RS_i scf Gas

2.4.1 Subsurface Sampling

In this case a subsurface sampler is lowered into the well and kept opposite the producing layer for a sufficiently long time, Figure 1. Many types of bottom hole samplers have been devised and described in the literature.

Subsurface samples can only be representative of the reservoir contents when the pressure at the point of sampling is above or equal to the saturation pressure. If this condition is not fulfilled, one should take a surface sample. Even at pressures close to the saturation pressure there is a serious possibility of sample integrity being lost as a result of the system going two phase during transfer to the sample chamber.

A simple test can be carried out in the field at the well site to find out whether a reliable sample has been obtained. In this test the sample cylinder is pressurised either by using a piston cell or using mercury as the displacing fluid. Whereas mercury was the most common fluid to be used as a pressure transfer and volume change fluid, because of toxic and other

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concerns its use is diminishing. From the relation between injection pressure and volume of mercury injected, the following properties are derived:

- (1) the pressure existing within the sampler when it is received at the surface;
- (2) the compressibility of the material within the sampler;
- (3) the bubble-point pressure of the contents of the sampler.

If two or three samples taken at short time intervals show the same measured properties, it is highly probable that good samples have been obtained.

In recent years there have been considerable advances on downhole fluid sampling.

The contents of the sampler are transferred by means of a mercury pump into a high pressure shipping container. Two fillings of a sampler of 600 ml capacity are usually sufficient for a complete PVT analysis. In recent years mercury has been replaced using the application of piston cells by some companies.

Subsurface sampling is generally not recommended for gas-condensate reservoirs nor for oil reservoirs containing substantial quantities of water.

Figure 2.9: Subsurface Sampling

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Fig. 2.10: Subsurface collection of PVT sample

2.4.2 Surface recombination sampling

In collecting fluid samples at the surface, separate volumes of oil and gas are taken at separator conditions and recombined to give a composite fluid sample. The surface equipment is shown schematically in fig. 2.11.

Fig. 2.11: Collection of a PVT sample by surface recombination

The well is produced at a steady rate for a period of several hours and the gas oil ratio is measured in scf of separator gas per stock tank barrel of oil. If this ratio is steady during the

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period of measurement then one can feel confident that recombining the oil and gas in the same ratio will yield a representative composite sample of the reservoir fluid. In fact, a slight adjustment must be made to determine the actual ratio in which the samples should be recombined. This is because, as shown in fig. 2.11, the oil sample is collected at separator pressure and temperature whereas the gas oil ratio is measured relative to the stock tank barrel, thus the required recombination ratio is

Dimensionally, the measured gas oil ratio must be multiplied by the shrinkage factor from separator to stock tank conditions. This factor is usually determined in the laboratory as the first stage of a PVT analysis of a surface recombination sample by placing a small volume of the oil sample in a cell at the appropriate separator conditions and discharging it (flash expansion) to a second cell maintained at the field stock tank conditions. During this process some gas will be liberated from the separator sample, due to the reduction in pressure and temperature, and the diminished stock tank oil volume is measured, thus allowing the direct calculation of S . In order to be able to perform such an experiment it is important that the engineer should accurately measure the pressure and temperature prevailing at both separator and stock tank during sampling and provide the laboratory with these data.

One of the attractive features of surface recombination sampling is that statistically it gives a reliable value of the producing gas oil ratio measured over a period of hours; furthermore, it enables the collection of large fluid samples. Of course, just as for subsurface sampling, the surface recombination method will only provide the correct gas oil ratio if the pressure in the vicinity of the well is at or above bubble point pressure. If not, the surface gas oil ratio will be too low or too high, depending upon whether the free gas saturation in the reservoir is below or above the critical saturation at which gas will start to flow. In this respect it should be emphasized that PVT samples should be taken as early as possible in the producing life of the field to facilitate the collection of samples in which the oil and gas are combined in the correct ratio.

2.5 DETERMINATION OF THE BASIC PVT PARAMETERS IN THE LABORATORY AND CONVERSION FOR FIELD OPERATING CONDITIONS

2.5.1 PVT APPARATUS

The apparatus required for PVT analysis consists of:

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- (a) apparatus for the transfer and the recombination of separator oil and gas samples;
- (b) apparatus for measuring gas-volumes and for performing separator tests;
- (c) the PV cell and displacement pumps and dispensing cell;
- (d) high pressure viscometer;
- (e) gas chromatograph.

Diagrams of the layouts for different systems are given in Figures 2.12.

Figure 2.12: Equipment for the Study of Subsurface Samples

2.5.2 RECOMBINATION OF THE SURFACE OIL AND GAS SAMPLES

The GOR given by the field refers to the separator tank gas-oil ratio. In order to recombine the separator gas and the separator oil in the correct ratio, the volumetric ratio between tank oil and separator oil is determined in the laboratory 85% of the cylinder containing separator oil is occupied by oil. There is a gas cap on top. This precaution is taken in view of the long transport time and the risk of great fluctuations in temperature involved.

After the gas cap has been dissolved by pressing water into the cylinder at a pressure higher than the separator pressure a given quantity of oil is then flashed at pressure and room temperature through a separator operating under the same conditions as the tank in the field i.e. at the same pressure and temperature. The collected tank oil is weighed and the

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density determined, after which the volume of oil is known. The volume and the density of the liberated gas (tank gas) are also determined. From the above measurements, the tank gas/tank oil ratio is also known. If the volume of tank oil is lower than the corresponding quantity of separator oil we speak of shrinkage, in the opposite case of expansion; shrinkage occurs when a large quantity of gas is produced and expansion occurs when a small quantity of gas is dissolved in the separator oil. When the shrinkage or the expansion is known, recombination can take place on a tank-oil basis.

2.5.3 PVT TESTS

At this stage we should remind ourselves of the main applications of the PVT data.

The three main application areas are;

- to provide data for reservoir calculations,
- to provide physical property data for well flow calculations
- for surface facility design.

Although all are cited as users of the data, the reservoir calculation requirement has provided the main driving force for the tests to be carried out. In surface facility design for example the more simplistic black oil approach around which the PVT analysis is structured is considered too limiting, and the main data for this application is the compositional analysis of the fluids.

Over recent years, as the data is subsequently applied to computer based simulation tools, the ability to handle more complex descriptions of the fluids has led to more extended compositional analysis, beyond the C7+ limit which was the basis for many years. It is common practice in some PVT laboratories to measure co-position to C28 and then define a C29+ component

In reservoir calculations the PVT tests and subsequent report provides the source of the reservoir engineering properties necessary to describe the behaviour of the reservoir over its development and production. The tests conducted therefore have to take into consideration the processes going on both above and below the saturation pressure.

There are four main PVT tests for oil systems plus associated compositional analysis:

1. Flash vaporisation or relative volume tests: Constant composition expansion of the

undersaturated oil from initial pressure to the bubble point.

2. Differential test: Differential vaporization: expansion below the bubble point with varying fluid composition.

3. Separator tests: Separator flash expansion experiments.

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4. Viscosity measurements: Measurement of oil and gas viscosities as functions of pressure.

5. Compositional measurements: Compositional analysis of the reservoir fluid.

All these experiments are performed under isothermal conditions at the reservoir temperature.

A simple layout of a PVT facility and PV cell with its associated equipment is given in figure 2.13 and 2.14 respectively.

Figure 2.13: Simple Schematic of PVT Facility for Oil Tests

Fig. 2.14: Schematic of PV cell and associated equipment

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2.5.3.1: Flash Expansion Experiment

A volume of undersaturated oil is expanded in a PV-cell in which all of the fluid is contained within the cell during a series of pressure decrements: usually from the initial pressure (or above) to the bubble point. After each pressure step, the oil volume is measured and reported as a relative volume (RV):

□

$$\square = (\square / \square)$$

□

in which V_b is the volume at the bubble point. The RV is necessarily less than unity and decreases with increasing pressure. Oil density and compressibility are also reported. The bubble point pressure is manifest, as described above, by the sudden large change in volume for a small change in pressure, due to the evolution of high compressibility gas.

2.5.3.2: Differential Liberation Experiment

The differential vaporization experiment defines the changing FVF and solution GOR below the bubble point pressure. Gas saturated oil is charged to a PV-cell and the pressure is reduced in steps below the bubble point; after each, the evolved gas is removed from the cell and its volume measured along with the volume of gas saturated oil remaining in the cell. Once the pressure has been reduced to atmospheric, the temperature is decreased from the reservoir value to 60°F meaning that the final state is the stock tank condition (14.7 psia, 60°F) at which the remaining oil in the cell is referred to as the "residual" volume (stb), to which all oil volumes are related at any stage of depletion.

From the volumetric measurements of the oil and gas it is possible to calculate:

B_{od} = FVF of the oil at each stage of depletion (rb/stb).

R_{Sd} = corresponding solution GOR (scf/stb).

The subscript "d" relates to the differential experiment and the initial values of these parameters at the bubble point are B_{obd} (rbbl/stb) and R_{sid} (scf/stb) respectively.

NOTE:

The main difference between the two types of experiment shown in fig. 2.15 (a) and (b) is that in the flash expansion no gas is removed from the PV cell but instead remains in equilibrium with the oil. As a result, the overall hydrocarbon composition in the cell remains unchanged. In the differential liberation experiment, however, at each stage of depletion the liberated gas is physically removed from contact with the oil and therefore, there is a continual compositional change in the PV cell, the remaining hydrocarbons becoming

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progressively richer in the heavier components, and the average molecular weight thus increasing.

Fig. 2.15: Illustrating the difference between (a) flash expansion, and (b) differential liberation

2.5.3.3: Separator Tests

The object of these tests are to examine the influence of separator pressure and temperature on formation volume factor, gas/oil ratio, gas density and tank-oil density. These tests are not driven by those who will be responsible for the optimised separation process if the field ultimately is developed. They are carried out to give an indication of the oil shrinkage and GOR which occurs when the fluids are produced to surface conditions. It should be emphasised at this stage that there is not a unique value for the formation volume factor and solution gas-oil ratio. It depends on the stages and conditions of separation through which the fluids pass.

With the equipment available, a single test or a multiple separation test can be carried out. The systems is set up as shown in the schematic below, Figure 2.16.

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Figure 2.16: Schematic of a Two Stage Separator Test

The procedure for the separation test is as follows. The starting point is oil in the PVT cell at its reservoir bubble point, i.e. the same starting condition as the differential test. Fluid is displaced from the PVT cell ensuring that the PVT cell contents remain at bubble point pressure. The gas and liquids are collected from the separation stage(s) and their respective properties measured. The final stage is at stock tank conditions. The resulting fluid is termed stock tank oil.

A single separator test is carried out by flashing reservoir liquid at bubble point pressure and reservoir temperature through the separator operating at the average annual temperature and at pressures which may be expected in the field. The difference in results when using a single or a double separator is that in the former case the total gas/oil ratio is higher, the shrinkage is greater and the density of the tank oil is higher than in the latter case.

The main objectives of the separator test are in combination with the flash vaporisation and differential tests to provide formation volume factor and solution gas-oil ratios over a full pressure range above and below the bubble point. In quoting these values it is important to recognise that the values are separator condition specific.

2.5.3.4: Viscosity

The viscosity is measured at reservoir temperature and at different pressures both above and below saturation pressure. It is important for viscosity measurements below the bubble point to generate the fluid for study by a differential mode to simulate the nature of the fluid that would exist at these conditions.

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Viscosity measurements were largely carried out with a “rolling ball” high pressure viscometer consisting of a highly polished-steel capillary of 1/4" dia. which can be closed at the top by means of a plunger and is provided at the bottom with a contact which is connected with an amplifier. A steel or platinum ball rolls in the capillary, its diameter hence slightly smaller than that of the capillary. When the ball reaches the bottom, it makes contact between the wall of the capillary and the point of contact, as a result of which a circuit is closed and a whistling sound is heard. The time of rolling is a measure of the viscosity.

In recent years the pressure drop along a capillary tube of known length and internal dimensions has been used. The viscosity being calculated using the Poiseuille equation, the laminar flow pressure drop equation for a pipe of a particular diameter and length.

Although being used it is clearly restricted by operating under a fixed flow regime, laminar and velocities to ensure that there is a sufficient pressure drop to measure and not too large to influence physical properties.

2.5.3.5: Hydrocarbon Analysis

A hydrocarbon analysis is made of the methane to an upper paraffin fraction of the recombined surface (or subsurface) sample. Historically this upper limit was C6 and the remainder lumped as a C7+ fraction. C10. Higher C numbers are now used as analytical methods enable even higher levels of characterisation. The plus component having been separated in a distillation column needs to be characterised by its specific gravity and its apparent molecular weight. The latter is achieved using a depression of freezing point method. The different components are determined by means of gas/liquid chromatography. If the description is based solely on paraffins then non paraffinic components are added to the next higher paraffin.

The value of a higher characterisation is particularly helpful to process engineering considerations, say up to C30 because at lower temperatures long chain hydrocarbons will form a solid phase (such as wax) and adhere to surfaces. At reservoir conditions they are in the liquid phase and therefore do not affect the reservoir flow process.

In some cases the fluids produced in the final separation stage are identified to a higher C

number together with aromatic and naphtha components. Compositions are also made of the produced gases from the various tests.

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2.5.4 COMPLETE PVT ANALYSIS

The complete PVT analysis for oil, provided by most laboratories, usually consists of the following experiments and calculations.

a) Compositional analysis of the separator oil and gas, for samples collected at the surface, together with physical recombination or compositional analysis of the reservoir fluid collected in a subsurface sample.

Such analyses usually give the mole fractions of each component up to the hexanes. The hexanes and heavier components are grouped together, and the average molecular weight and density of the latter are determined.

b) Flash expansion conducted at reservoir temperature. This experiment determines

- the bubble point pressure
- the compressibility of the undersaturated oil
- the total volume v_t of the oil and gas at each stage of depletion.

c) Differential liberation experiment to determine

- E , Z , F and v_o , with F and v_o measured relative to the unit volume of bubble point oil.

Alternatively, by measuring $c_b d$ during the last stage of the differential liberation, the above data can be presented as

- E , Z , $R_{si d} - R_{s d}$ (or just $R_{s d}$) and $B_o d$, with $R_{s d}$ and $B_o d$ measured relative to residual oil volume. In addition, the gas gravity is measured at each stage of depletion.

d) Measurement of the oil viscosity at reservoir temperature (generally using the rolling ball viscometer), over the entire range of pressure steps from above bubble point to atmospheric pressure. Gas viscosities are normally calculated at reservoir temperature, from a knowledge of the gas gravity, using standard correlations.

e) Separator tests to determine the shrinkage, $c_b f$, and solution gas oil ratio, $R_{si f}$, of unit volume of bubble point oil (1 barrel) when flashed through various separator combinations. Instead of actually performing these tests, in many cases the results are obtained using the phase equilibrium calculation technique.

f) Composition and gravity of the separator gas in the above separator tests.

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2.5.5 COMPARISON OF LABORATORY AND FIELD PVT DATA

No matter how careful the engineer may be in collecting fluid samples for laboratory PVT analysis, it seldom happens that the resulting functions match exactly the P W relations demonstrated by the field itself, once it has been brought on continuous production. There can be many reasons for the disparities, such as deferring the sampling until too late in the

test or possibly because the separators installed are not quite as theoretically planned.

But perhaps the main reason is simply that if a well is sampled at the appraisal stage during the course of a DST, it must be appreciated that it is producing in a non-equilibrium state, especially with respect to conditions in the vicinity of the wellbore and the collection of completely representative samples in a few "small bottles" is too much to be expected.

Although laboratory techniques are very sophisticated, the experimental PVT results are only as reliable as the fluid samples collected and should never be regarded as more than a good approximation to the real fluid properties. Laboratory PVT results must be calibrated and, as is often necessary, corrected to match field observations after the start of production. At this time, the engineer should monitor the production data very carefully and, in particular, the producing GOR through the separator configuration used in the field and seek any evidence that the reservoir has passed through the bubble point pressure, which will occur in depletion type fields.

The typical GOR behaviour of fractured and homogeneous acting reservoirs is depicted in Fig. 2.7a. In the former, there is usually little or no development of a critical gas saturation, S_{gc} , below the bubble point, which must be exceeded before the liberated gas becomes mobile. Instead the free gas tends to flow towards the wellbore through the fracture system as soon as it is released. In this case the saturation pressure can be directly related to the prevailing reservoir pressure at which the GOR begins to rise sharply above the value of, RS_i defined during the initial phase of production at undersaturated conditions (Fig. 2.7a).

Fig. 2.7. (a) GOR behaviour of fractured and homogeneous reservoirs. (b) Correction of laboratory measured GOR for field conditions.

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In the case of a more homogeneous reservoir, the development of a critical gas saturation is usually observed below the bubble point and is manifest as a slight decrease in the producing GOR as the small volume of gas is trapped in the reservoir rather than produced to the surface (Fig. 2.7a). The bubble point can then be defined as the pressure at which the decline in GOR first occurs, although this is often a more subtle observation than for fractured reservoirs. Other ways of confirming that the reservoir pressure has fallen below the bubble point are by observation of:

- an increase in the API gravity of the oil
- noting any sudden reduction in the productivity of wells

If it is determined, by direct observation of reservoir performance, that the initial GOR is different from that determined in the PVT analysis, then it becomes necessary to correct the laboratory PVT functions. If, as shown in Fig. 2.7b, the field GOR exceeds that determined by experiment and the elevated bubble point pressure has also been observed, then the RS trend below the bubble point can simply be "eye-balled" in the knowledge that at atmospheric pressure $p \rightarrow 0$. A more scientific approach might be to use standard correlations to predict the decline or even more sophisticated EOS thermodynamic software packages. If the bubble point has not been directly observed the latter techniques must be

applied to determine it and, whether the bubble point has been observed or not, they need to be used to correct the oil FVF both above and below the bubble point pressure.

The PVT properties of reservoir fluids can be somewhat more complex than determined in the laboratory by performing experiments on samples collected at the appraisal stage of field development and can vary throughout the lifetime of a project. Consequently, the reservoir engineer should take advice from specialised process engineers before finalising the input to studies, whether analytical or using numerical simulation.

Comparison of laboratory and field parameters for the differential liberation experiment is shown in the table below:

Table 2.1: Laboratory and field parameters of differential liberation experiment

Laboratory Differential Field Parameter Conversion

Parameter

ρ_{lab} ρ_{field}

$\rho_{\text{lab}} = (\rho_{\text{lab}}/\rho_{\text{field}})$

ρ_{lab}

F ρ_{lab} ρ_{field}

$\rho_{\text{lab}} = \rho_{\text{field}} - 5.615 (\rho_{\text{lab}}/\rho_{\text{field}})$

ρ_{lab}

E ρ_{lab} 1

$\rho_{\text{lab}} = (\rho_{\text{lab}}/\rho_{\text{field}})$

5.615 ρ_{lab}

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2.6 CALCULATION OF HYDROCARBON VOLUMES

The volume of oil and gas in-place depends on a number of parameters :

- The aerial coverage of the reservoir, A
- The thickness of the reservoir rock contributing to the hydrocarbon volume. h_n
- The pore volume, as expressed by the porosity ϕ , the reservoir quality rock.
- The proportion of pore space occupied by the hydrocarbon (the saturation), $1-S_w$

Oil Initially In Place (OIIP)

Consider a reservoir which is initially filled with liquid oil. The oil volume in the reservoir (oil in place) is

$$V_{\text{OIIP}} = A \phi (1 - S_{\text{wi}}) = h_n A \phi (1 - S_{\text{wi}}) (\rho_{\text{oi}})$$

Where:

A= average area

h_n = nett thickness. Net thickness = gross thickness x net: gross ratio

V = the net bulk volume of the reservoir rock

ϕ = the porosity, or volume fraction of the rock which is porous, and

S_{wi} = the connate or irreducible water saturation and is expressed as a fraction of the pore

volume.

The product V_{ϕ} is called the pore volume (PV) and is the total volume in the reservoir which can be occupied by fluids. Similarly, the product $V_{\phi} (1 - S_{wc})$ is called the hydrocarbon pore volume (HCPV) and is the total reservoir volume which can be filled with hydrocarbons either oil, gas or both.

Stock Tank Oil Initially In Place (STOIIP)

To convert volumes at reservoir conditions to stock tank conditions, formation volume factors are required where B_O are the oil formation volume factor. The expression of original oil in place is termed the STOIIP. When expressed as stock tank volumes, equation above is divided by the formation volume factor B_O

$$V_{STOIIP} (1 - S_{wc})$$

$$V_{STOIIP} = (V_{\phi} B_O)$$

$$V_{STOIIP}$$

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Recoverable reserves

The recovery factor, RF, indicates the proportion of the in-place hydrocarbons expected to be recovered. To convert in place volumes to reserves we need to multiply the STOIIP by the recovery factor. The equation describing this method of applying equity is:

$$V_{\phi} (1 - S_{wc})$$

$$V_{\phi} (1 - S_{wc}) \times RF$$

$$V_{\phi}$$

$$\text{Reserves} = \text{STOIIP} \times RF$$

Movable oil

This is defined as the oil that can be physically involved during a secondary recovery flood by either water or gas injection. The equity equation can be expressed as:

$$V_{\phi} (1 - S_{wc} - S_{or})$$

$$V_{\phi} (1 - S_{wc} - S_{or}) =$$

$$V_{\phi}$$

where S_{or} = residual oil saturation

This is a fairly rare form of organising a determination and gives rise to the complication of measuring and mapping the residual oil saturation throughout the reservoirs. It requires heavy involvement from the reservoir engineers since the measurement of residual oil saturation in waterflooding experiments on thin core plugs is largely regarded as their domain

2.7 CALCULATION OF GAS INITIALLY IN PLACE

The important parameter required in the calculation of the gas initially in place (GIIP) is the gas expansion factor, which at initial conditions is:

$$Z$$

$$\alpha = 35.37 \left(\frac{p}{p_r} \right)$$

$$\alpha =$$

in which α is the dimensionless Z-factor accounting for molecular scale effects which must be accounted for in the PVT for gas at reservoir conditions. T is the absolute temperature in degrees Rankine (= OF + 460). The gas expansion factor relates surface to reservoir volumes of gas and is related to the gas formation volume factor as:

$$1$$

$$\alpha = \left(\frac{p}{p_r} \right)$$

$$5.615 \alpha$$

The GIIP can then be calculated as:

$$G_{IIP} = \alpha \phi (1 - \alpha) N (V_{OC})$$

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3.0 MATERIAL BALANCE EQUATION

3.1 INTRODUCTION

The Schilthuis material balance equation has long been regarded as one of the basic tools of reservoir engineers for interpreting and predicting reservoir performance. In this section, the zero dimensional material balance is derived and subsequently applied, using mainly the interpretative technique of Havlena and Odeh, to gain an understanding of reservoir drive mechanisms under primary recovery conditions.

Although the classical material balance techniques, once applied, have now largely been superseded by numerical simulators, which are essentially multi-dimensional, multi-phase, dynamic material balance programs, the classical approach is well worth studying since it provides a valuable insight into the behaviour of hydrocarbon reservoirs.

3.2 DERIVATION OF THE CUMULATIVE MATERIAL BALANCE FOR OIL RESERVOIRS

The derivation of the volumetric material balance equation was first presented in 1936 by Schilthuis. To appreciate the full range of complexity in derivation, consider the case of depletion of the reservoir depicted in Fig. 3.1, which has an active aquifer and a gas cap.

Fig. 3.1. Reservoir considered in the derivation of the fully expanded material balance equation.

If N is the STOIIP in the oil column and

HCPV of the gascap

$$\alpha =$$

HCPV of the oil column

at initial conditions, which is therefore a constant, then applying the defining expression for the STOIIP,

$$G_{IIP} = N = \alpha \phi (1 - \alpha) N (V_{OC})$$

and

$$G_{IIP} = N (V_{OC})$$

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giving a total HCPV of $(1 + m)NBO_i$ (rb).

Suppose that at a given time after the start of production the pressure has dropped from its initial value at datum, p_i (psia), to some current average value, p (psia), as the result of fluid withdrawal from the reservoir. Then, if during this pressure drop, Δp , the entire reservoir system is allowed to expand (in an artificial sense) underground, the total expansion plus any natural water influx must equal the volume of fluids expelled from the reservoir as production. The volumetric material balance can then be expressed in reservoir barrels as:

$$\text{Volume of fluids produced} = \text{Volume of reservoir expansion} + \text{Volume of water influx}$$

Perhaps the simplest way to visualise the balance is that if a measured surface volume of production (oil/gas/water) were returned to the reservoir at the reduced pressure, p , it must fit exactly into the volume which is the sum of the total expansion and the water influx. The material balance equation can then be evaluated as follows.

3.2.1 Left-hand side (underground withdrawal - rb)

Cumulative oil production: The reservoir depicted in Fig. 3.1 is particularly complicated in that it contains a gas cap. Consequently, the initial pressure in the oil column must also be equal to the bubble point pressure, $p_i = p_b$, since the oil is gas saturated; therefore, even a slight drop in the pressure will cause the liberation of solution gas in the reservoir. The cumulative volumes of produced fluids measured at the surface are:

N_p = cumulative oil (stb)

W_p = cumulative water (stb)

G_p = cumulative gas (scf)

and:

$N_p = N_p \text{ (stb)}$

$W_p = W_p \text{ (stb)}$

$G_p = G_p \text{ (scf)}$

which is the cumulative or average GOR since the start of production.

Returning the produced fluids to the reservoir at the reduced pressure, p , at which the FVF's and solution GOR are: BO (rb/stb), BW (rb/stb), B_g (rb/scf) and RS (scf/stb) gives:

oil + dissolved gas = $N_p BO$ (rb)

water = $W_p BW$ (rb)

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free gas = $G_p (1 - RS) BW$ (rb)

The first of these contains $N_p BO$ scf of dissolved gas, consequently the free gas to be returned is $G_p (1 - RS) BW$ (scf). To complicate matters, some of the free gas will be that evolved from the oil column in producing below the bubble point while there may also be a component of gas cap gas which is eventually produced; the difficulty is distinguishing between the two. The cumulative underground withdrawal is therefore:

$$V_o [(1 + (C_o - C_{oi}) \Delta p)] + V_{oi} C_{oi} \Delta p \quad (rb) \quad 3.2$$

3.2.2 Right-hand side (expansion plus water influx)

In allowing the total system to expand as the pressure drops from p_i to p , consideration must be given to everything that can change volume during the depletion: the oil and its originally dissolved gas, the gas cap, the connate water and the reservoir pore volume; added to which is the cumulative water influx. The separate components may be accounted for as follows.

1) Oil plus originally dissolved gas: The original HCPV of the oil column is V_{oi} (rb). Since the reservoir under study, Fig. 3.1, is initially at the bubble point, reducing the pressure will result in the release of gas and the shrinkage of oil such that its change in volume is:

$$N(C_{oi} - C_o) \Delta p \quad (rb) \quad 3.3$$

The initial volume of gas in the oil column is V_{gi} (scf) and the amount still dissolved at the reduced pressure, p , is V_g (scf). Therefore, the amount liberated is:

$$N(V_{gi} - V_g) \Delta p \quad (rb) \quad 3.4$$

and the total change in volume (expansion) of the oil column is:

$$N[(C_{oi} - C_o) + (V_{gi} - V_g) \Delta p] \quad (rb) \quad 3.5$$

2) Gas cap: The original HCPV of the gas cap is $mNBO_i$ (rb) which expressed as a surface volume is:

$$V_{gi}$$

$$V_{gi} (rb)$$

$$V_{gi}$$

and, at the reduced pressure, p , it occupies a reservoir volume of:

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$$V_g$$

$$V_{gi} (rb)$$

$$V_{gi}$$

The gas cap expansion is therefore:

$$V_g$$

$$V_{gi} (1 - C_o) \Delta p \quad (rb) \quad 3.6$$

$$V_{gi}$$

3) Connate water: Although of low compressibility, the volume of connate water in the reservoir is large and therefore its expansion must be catered for during depletion. If C_{wi} is the average connate water saturation, determined as described in section 3.4, then its expansion can be calculated applying the definition of isothermal compressibility to the water as:

$$V_{wi} = V_o C_{wi} \Delta p$$

where V_o is the total volume of water:

$$\frac{V_{p0}}{V_{p0} - V_{p1}} = \frac{V_{p0}}{V_{p0} - V_{p1}}$$

$$1 - \frac{V_{p1}}{V_{p0}} = \frac{V_{p0} - V_{p1}}{V_{p0}}$$

$$1 - \frac{V_{p1}}{V_{p0}} = \frac{V_{p0} - V_{p1}}{V_{p0}}$$

The total HCPV, as defined above, is $(1 + \beta)V_{p0}$ (rb), therefore:

$$(1 + \beta)V_{p0} = V_{p0} + \beta V_{p0}$$

$$V_{p0} = (1 + \beta)V_{p0}$$

$$1 - \frac{V_{p1}}{V_{p0}}$$

4) Pore compaction: As fluids are produced and the pressure declines the entire reservoir pore volume is reduced (compaction) and the change in volume, although negative, expels an equal volume of fluid as production and is therefore additive to the expansion terms. Again, applying the definition of isothermal compressibility to the entire pore volume:

$$(1 + \beta)V_{p0} = V_{p0} + \beta V_{p0}$$

$$\beta(V_{p0}) = \frac{V_{p0} - V_{p1}}{V_{p0}} = \beta(V_{p0})$$

$$1 - \frac{V_{p1}}{V_{p0}}$$

5) Water influx: If the reservoir is connected to an active aquifer then, once the pressure drop is communicated either wholly or partially throughout its volume, the water will expand resulting in an influx of W_e (stb) or W_e (rb).

Adding equations 3.5, 3.6, 3.7 and 3.8, together with the influx and equating the sum to the underground withdrawal, equation 3.2, gives an expression for the Fully Expanded Material Balance as:

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$$V_{p0} \left[\frac{V_{p0} - V_{p1}}{V_{p0}} + \frac{V_{p0} - V_{p1}}{V_{p0}} \right] + V_{p0} \left[\frac{V_{p0} - V_{p1}}{V_{p0}} \right]$$

$$V_{p0}$$

$$= N[(V_{p0} - V_{p1}) + (V_{p0} - V_{p1}) + (V_{p0} - V_{p1})] + V_{p0} (1 - 1)$$

$$V_{p0}$$

$$(1 + \beta)V_{p0} (V_{p0} + V_{p1}) \Delta p$$

$$+ + V_{p0} (3.9)$$

$$1 - \frac{V_{p1}}{V_{p0}}$$

In this general form there are so many unknowns in the equation (section 3.4) that the attainment of an exact or unique solution is virtually impossible. Fortunately, most reservoirs are simpler than that depicted in Fig. 3.1, leading to more straightforward formulations of the equation. In terms of importance it is as powerful in reservoir engineering as Einstein's $E = mc^2$ in nuclear physics but unfortunately is a little more cumbersome in form.

If there is water or gas injection, or both, into the reservoir, the cumulative surface volumes: W_e (stb) and W_e (scf) would represent equivalent reservoir volumes at pressure p of V_{p0} (rb) and V_{p0} (rb), in which V_{p0} is the FVF of the lean injected gas and it is assumed that the water is injected free of gas. The cumulative injected volumes can be either subtracted from the left-hand side of the material balance, resulting in a net underground withdrawal, or

added to the expansion terms on the right, as appropriate in the particular application.

3.3 THE MATERIAL BALANCE EXPRESSED AS A LINEAR EQUATION

Since the advent of sophisticated numerical reservoir simulation techniques, the Schilthuis material balance equation has been regarded by many engineers as being of historical interest only; a technique used back in the nineteen forties and fifties when people still used slide-rules. It is therefore interesting to note that as late as 1963-4, Havlena and Odeh presented two of the most interesting papers ever published on the subject of applying the material balance equation and interpreting the results. Their papers described the technique of interpreting the material balance as the equation of a straight line.

In an attempt to condense the material balance into a more tractable form, the nomenclature of Havlena and Odeh will be employed throughout this section. This reduces equation 3.9 to the form:

$$N_p = N(\alpha + \beta + \gamma) + N_0 \quad 3.10$$

in which:

$$\alpha = N[\alpha + (\beta - \alpha)N] + N_0 N(\alpha) \quad 3.11$$

$$\beta = [(\alpha - \alpha_0) + (\alpha_0 - \alpha)N] (N/N_0) \quad 3.12$$

$$\gamma =$$

$$N = N_0 (1 - \alpha/N_0) \quad 3.13$$

$$N_0 =$$

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$$(N_p N_0 + N) \Delta$$

$$N_0 = (1 + \alpha) N_0 (N/N_0) \quad 3.14$$

$$1 - \alpha$$

The final three terms, having units of rb/stb, represent underground expansions per unit volume of stock tank oil.

3.4. SOLVING THE MATERIAL BALANCE (KNOWN AND UNKNOWN)

Considering the fully expanded material balance equation, 3.9, the parameters involved can be divided into the following categories:

Should be "known" Potential unknowns

NP N

RP We

WP p

cw BO, RS, Bg

SWC m

BW cf

Total 6 Total 8

The score of 6 knowns and 8 potential unknowns reveals the perennial difficulty not just

with material balance but throughout the subject of reservoir engineering as a whole: there are far too few equations for the number of unknowns for which they must be solved. Numerical simulation does not alleviate the condition, it merely adds to the potential unknowns appearing in the sparse equations: geometry, ϕ , k , k_r , etc. But rather than writing off reservoir engineering as a serious subject, the uncertainty simply adds to its allure, throwing emphasis on observation and judgement rather than numeracy for there can be rarely a unique mathematical solution to problems. Considering further the "knowns" and "unknowns."

Knowns: For strictly commercial reasons, NP (cumulative oil production) is the best "known" of all but, frustratingly, in many older fields or fields that are in remote locations where produced gas cannot be utilised, RP (cumulative or average GOR = GP/NP) and WP (cumulative water production) are not necessarily measured. This moves them into the unknown category and exacerbates the difficulty in making sense of the material balance, if not making it an impossibility. It is assumed that the petrophysical evaluation is "always correct", in which case the average connate water saturation for the reservoir is evaluated as:

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$$\bar{S}_{wi} = \frac{\sum (h_i \phi_i S_{wi})}{\sum h_i \phi_i} \quad 3.15$$

which is averaged over all the wells and the subscript "i" relates to the different layers selected within each.

Unknowns: From early in the appraisal stage of a field there is always a volumetric estimate of the STOIP, N , available but as soon as there is production/pressure history for a reservoir this figure must be disregarded and an attempt made to establish it through solution of the material balance. The value so determined is the "effective" or "active" STOIP: that oil contributing to the production/pressure history, which is usually somewhat different from the volumetric estimate (usually being smaller) due to oil being trapped in undrained fault compartments or low-permeability regions of the reservoir.

3.5. COMPARISON BETWEEN MATERIAL BALANCE AND NUMERICAL SIMULATION MODELLING

It is a quite widely held belief that since the advent of numerical simulation modelling in the 1960's, material balance can now be discarded as an approximate technique of historical interest only. Nothing could be further from the truth, however, and anyone who embarks on a numerical simulation study without first applying material balance to the observed production/pressure history of a field is running the risk of error.

Fig. 3.4. Contrast between material balance and numerical simulation in history matching field performance.

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Consider an undersaturated oil reservoir in which the principle unknowns are the STOIP, N , and whether there has been a finite water influx, W_e , or not. The difference between

material balance and simulation in matching field performance for such a reservoir is illustrated in Fig. 3.4.

Provided the PVT and pressure decline trend have been rationalised then they are input to material balance calculations along with the production data as known quantities, nothing else is required. Then by "adept manipulation" - as described in later sections of the chapter - an attempt is made to solve the material balance equation to determine the two principle unknowns, N and We , the latter representing the drive mechanism. The main advantage in this approach is that provided an average pressure decline can be defined for the system, then no geometrical model is required. The accumulation is seen simply as an underground "blob" of hydrocarbons. In this respect the model is zero dimensional with the average pressure determined at a point (datum plane) which is representative of the reservoir as a whole. It is this unique property that permits the attempted solution of the equation to determine the hydrocarbons in place and define the drive mechanism.

The alternative approach of employing numerical simulation modelling to match production performance is also illustrated in Fig. 3.4. Now the input must contain a full geological and petrophysical description of the reservoir necessary to give physical structure to the model. But if these are defined then so too is the STOIP which is therefore input as a known quantity. Similarly a part of the physical description is the aquifer model, one is either (mathematically) attached to the reservoir or it is not, at the discretion of the engineer. Consequently, the drive mechanism is also being fed in as an input assumption or partial assumption (i.e. if the aquifer is defined through history matching with the simulator then its properties are dependent on the assumed STOIP). Therefore, since the required answer, the STOIP, must necessarily be specified from the outset, then numerical simulation applied to the history matching of field performance can hardly be regarded as an investigative technique: the results achieved being simply a reflection of the engineer's input assumptions. The remainder of the simulation input consists of the production and PVT data together with SCAL results most notably the relative permeability functions required to shift fluids from one grid block to the next. Strangely enough the pressures are not input to the model as known quantities, instead individual grid block pressures and fluid saturations comprise the "solution" achieved at the end of each time step by simultaneously solving large numbers of second order differential equations expressed in finite difference form. In this respect the simulator tackles the problem backwards: "unknowns" such as the STOIP are fed in to determine the "knowns" such as the pressure.

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Paper 2 of 2 — CHE 525 FUNDAMENTALS OF RESERVOIR ENGINEERING

Level 500 • Second Semester • Handout

CHE 525

FUNDAMENTALS OF

RESERVOIR ENGINEERING

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NOMENCLATURE

SYMBOLS DESCRIPTION UNITS

A Area m^2

B Formation Volume Factor rb/stb

β_o Oil Formation Volume Factor rb/stb

β_g Gas Formation Volume Factor rcf/scf

β_w Water Formation Volume Factor rb/stb

C Isothermal Compressibility $1/\text{psi}$

α_p Pore Compressibility $1/\text{psi}$

α_o Oil Compressibility $1/\text{psi}$

α_w Water Compressibility $1/\text{psi}$

D Vertical (Reservoir) depth ft

e exponential

E_i exponential integral function

E Gas Expansion Factor scf/rcf

ϵ A term in material balance equation which connate H_2O expansion + pore compaction

ϵ_o Expansion + pore compaction = Expansion of oil

ϵ_g term in the material balance equation accounting for the expansion of the gas cap (rb/stb)

ϵ_o term in the material balance equation accounting for the expansion of the oil and its originally dissolved gas (rb/stb)

F underground withdrawal of fluids (rb)

G Gas Initially In Place scf

G_p Cumulative Gas production scf

h Formation thickness m

k Permeability mD

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L Length ft

m Ratio of initial hydrocarbon pore volume of the gas cap to that of the oil

column

m Slope of the early linear section of semi-log plot for well test analysis psi/log cycle

m End point mobility ratio number of moles

n Number of moles

N Stock Tank Oil Initially In Place stb

$\square\square$ Cumulative Oil Recovery stb

P Pressure Psia

$\square\square$ Bubble Point Pressure Psia

$\square\square$ Dimensionless Pressure

$\square\square$ Pressure @ external boundary Psia

$\square\square$ Initial Pressure Psia

$\square\square\square$ Pressure @ Standard Condition Psia

$\square\square\square$ Well bore flowing pressure Psia

$\square\square\square$ Well bore static pressure Psia

$\square$ Average Pressure Psia

q Liquid Production Rate bpd

$\square\square\square$ Water Injection Rate stb/d

Q Gas Production Rate mmscfpd

r Radial distance ft

$\square\square$ External Radius ft

$\square\square\square$ Aquifer/reservoir radius ratio ft

$\square\square$ Well bore radius ft

R Production Gas Oil Ratio scf/stb

R Universal gas constant J mol⁻¹ K

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$\square\square$ Cumulative Gas Oil Ratio Psia

S Skin factor

$\square\square$ Gas saturation

$\square\square\square$ Residual Gas saturation

$\square\square$ Oil saturation

$\square\square\square$ Residual oil saturation

$\square\square\square$ Connate Water saturation

t Time hr or yr.

T Temperature OR or K

U Aquifer constant

□ Velocity of flow ft/hr

V Volume ft³

w Width ft

□□ Dimensionless cumulative H₂O

□□ Cumulative water influx

□□ Cumulative water produced

Z Compressibility factor

ABBREVIATION

DST Drill Stem Test

RFT Repeat Formation Tester

BHP Bottom Hole Pressure

FVF Formation Volume Factor

GOR Gas Oil Rate

HCPV Hydrocarbon Pore Volume

PVT Pressure Volume Temperature

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1.0 INTRODUCTION TO RESERVOIR ENGINEERING

1.1 WHAT IS A RESERVOIR?

A reservoir is formed of one (or more) subsurface rock formations containing liquid and/or gaseous hydrocarbons, of usually sedimentary origin. The reservoir rock is porous and permeable, and the structure is bounded by impermeable barriers which traps the hydrocarbons. Figure 1.1 shows a cross section of a typical hydrocarbon reservoir.

Fig 1.1: Cross section of a typical hydrocarbon reservoir

1.2 WHAT IS RESERVOIR ENGINEERING?

- Reservoir engineering may be defined as the application of scientific principles to the drainage problems arising during the development and production of oil and gas reservoirs.
- It has also been defined as “the art of developing and producing oil and gas fluids in such a manner as to obtain a high economic recovery.
- Reservoir Engineering is a part of Petroleum Science that provides the technical basis for the recovery of petroleum fluids from subsurface sedimentary-rock reservoirs.

The Fig. 1.2 below indicates the place and role of Reservoir Engineering in the broad field of Petroleum Science

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Figure 1.2: Reservoir engineering and petroleum science.

1.3 ACTIVITIES IN RESERVOIR ENGINEERING

Reservoir engineering shares the distinction with geology in being one of the great "underground sciences" of the oil industry, attempting to describe what occurs in the wide open spaces of the reservoir between the sparse points of observation - the wells. In applying the subject, it is possible to define four major activities, which are:

- Observations
- Assumptions
- Calculations
- Development decisions

and these can be described as follows.

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1.3.1 Observations:

These include the geological model, the drilling of wells and the data acquired in each: cores, logs, tests, fluid samples. Following the start of field production, the oil, gas and water rates must be continuously and accurately monitored together with any injection of water and gas. Frequent pressure and production logging surveys should also be conducted throughout the lifetime of the project.

It is pointed out that the material balance equation can easily contain eight "unknowns" and failure to collect the essential rate, pressure and pressure-volume-temperature (PVT) data to give oneself a "sporting chance" of attaining a meaningful solution of the equation can make the application of quantitative engineering impossible.

Once the data have been collected and verified, the engineer must interpret them very carefully and collate them from well-to-well throughout the reservoir and adjoining aquifer. This is a most delicate phase of the whole business of understanding reservoirs, in which it can prove dangerous to rely too much on automated techniques. It is fine to use computer programs to interpret well tests, for instance, but to apply sophisticated numerical methods to generalise about formation properties by such means as generating statistical correlations and use of regression analysis should be reduced to a minimum.

1.3.2 Assumptions:

Now the main difficulty arises: having thoroughly examined and collated all the available data, the engineer is usually obliged to make a set of assumptions concerning the physical state of the "system" for which an appropriate mathematical description must be sought.

For instance:

- The oil or gas reservoir is or is not affected by natural water influx from an adjoining aquifer.

- There will or will not be complete pressure equilibrium across the reservoir section under depletion or water drive conditions.

'This is the crucial step in practical engineering. The third activity, calculations, is entirely dependent on the nature of the assumptions made as is the fourth, development decisions, which rely on the results of the calculations.

It is therefore necessary to be extremely cautious when making physical assumptions and the most convincing reservoir studies are those containing the least number. In this respect, the application of material balance is strongly recommended in history matching field performance, as being one of the safest techniques in the business for the simple reason that it requires fewer (far fewer) assumptions than the alternative technique, numerical

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simulation modelling, which should therefore be used as a logical follow-up. Assumptions can usually be confirmed only by observation, not calculation, but this can be problematical.

1.3.3 Calculations:

'While the above comments may seem to denigrate the importance of calculations, that is only meant to be relative to the physical assumptions upon which any mathematical model selected for use is entirely dependent. Once a physical condition has been defined (assumed) then calculations are an absolute must.

1.3.4 Development decisions:

Every action contemplated, planned and executed by reservoir engineers must lead to some form of development decision - otherwise it should not be undertaken in the first place.

1.4 TECHNICAL RESPONSIBILITIES OF THE RESERVOIR ENGINEER

Four distinct technical responsibilities can be defined:

- contributing, with the geologists and petrophysicists, to the estimation of hydrocarbons in place
- determining the fraction of discovered hydrocarbons that can be (reasonably) recovered
- attaching a time scale to the recovery
- day-to-day operational reservoir engineering throughout the lifetime of the project.

1.4.1 Appraisal

The reservoir engineer is involved in the development from the day of discovery of the accumulation and is (or should be) responsible for the collection, collation and interpretation of the following data in the discovery and subsequent appraisal wells.

- Well tests (DSTs)
- Pressure-depth relationships
- Fluid sampling
- Coring

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The most important information gathered from a well test (DST) is the type of hydrocarbon present and whether the formation capable of producing at what may be judged as a commercial rate.

One of the main results determined from pressure-depth surveys, best obtained using the RFT, is the location of fluid contacts (gas-oil, oil-water, gas-water contacts) which are required in the estimation of hydrocarbons in place. The determination of fluid pressure gradients required in such work depends on the collection of reliable fluid samples and their accurate PVT analysis in the laboratory. Also dependent on the PVT analysis is any form of recovery calculation performed and the design of surface facilities to handle separate and dispose of the produced fluids, the oil, gas and water.

Core data is one of the most important requirements from reservoirs in which it is intended that secondary recovery (water-oil or gas-oil) displacement will be the elected means of development. What matters in viewing core data is the all-important permeability distribution across the producing formations; it is this, more than anything else, that dictates the efficiency of displacement processes.

1.4.2 End of appraisal

Usually, the timing of this tends to be gauged by the explorers, when they feel satisfied that they have acquired sufficient data to define within a tolerable range of certainty whether they have proven sufficient stock tank oil initially in place gas initially in place for a viable field development. But this must not be regarded as the sole criterion upon which development decisions are made. The reservoir engineer must be also heavily involved concerning when the appraisal phase should be terminated, based on whether enough meaningful production orientated data have been acquired to perform long-term project development calculations to evaluate:

- Production profiles of oil, gas and water
- Injection profiles of water and/or gas
- Production/injection well]requirements
- Surface topsides facilities (design.

1.4.3 Project Development

This is regarded as the period extending from the time that continuous hydrocarbon production commences until the time of field abandonment. Throughout this time, some of the day-to-day activities of the field reservoir engineer are as follows:

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New well locations, Well completion intervals, Regular well surveys, Areal fluid movement, Production profiles, Decline policy, Enhanced oil recovery

1.5 PRINCIPLES OF RESERVOIR ENGINEERING

The working tools of the reservoir engineer are subsurface geology, applied mathematics, and the basic laws of physics and chemistry governing the behaviour of liquid and vapour phases of crude oil, natural gas, and water in reservoir rock.

To describe 90% of reservoir engineering (excluding thermal effects) requires application of the following physical principles:

- Conservation of mass
- Darcy's law
- Isothermal compressibility
- Newton's second and third laws of motion

1.5.1 Darcy's law

This states that if there is flow through a porous media, there is a relationship for the amount of flow. For one dimensional, linear, horizontal flow (core flooding experiment), Darcy's law (experimental law or empirical relationship) can be expressed in absolute units as:

$$q = - \frac{k A}{\mu L} \frac{dp}{dx}$$

$$q = - \frac{k A}{\mu L} \frac{dp}{dx}$$

$$q = - \frac{k A}{\mu L} \frac{dp}{dx}$$

or in field units as

$$q = - \frac{k A}{\mu L} \frac{dp}{dx}$$

$$q = -1.27 \times 10^{-3} \left(\frac{k A}{\mu L} \right) \frac{dp}{dx}$$

$$q = - \frac{k A}{\mu L} \frac{dp}{dx}$$

For linear flow, the pressure gradient dp/dx is intrinsically negative hence the minus sign to make the rate q positive. Note that the state or phase of a fluid in the reservoir usually changes with pressure, the temperature remaining substantially constant.

1.5.2 Isothermal compressibility

Isothermal compressibility accounts for the change in volume of fluids, and the pore volume itself, as the pressure varies at constant temperature. It is defined as:

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$$\beta = - \frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$$

$$\beta = - \frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$$

$$\beta = - \frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$$

Since the rate of change of volume with respect to pressure is negative, the minus sign is required to assure that the compressibility is positive.

1.5.3 Newton's second and third laws of motion

The second law defines force as equal to mass times acceleration and, in particular, defines the force resulting from the attraction of gravity. Newton's third law of motion states that "action and reaction are equal and opposite". When considering the displacement of one fluid by another with different density, it is imperative that the effect of gravity be

accounted for in displacement efficiency calculations in macroscopic reservoir sections. The realism of water displacing oil or gas displacing oil in the field is that the process occurs in hillsides rather than core plugs and in this domain gravity predominates.

1.6 OIL AND GAS RESOURCES AND RESERVES

Reserves are estimated volumes of crude oil, condensate, natural gas, natural gas liquids, and associated substances anticipated to be commercially recoverable from known accumulations from a given date forward, under existing economic conditions, by established operating practices, and under current government regulations. Reserve estimates are based on geologic and/or engineering data available at the time of estimate.

The relative degree of an estimated uncertainty is reflected by the categorisation of reserves as either "proved" or "unproved"

a) Proved Reserves can be estimated with reasonable certainty to be recoverable under current economic conditions. Current economic conditions include prices and costs prevailing at the time of the estimate. Reserves are considered proved if commercial producibility of the reservoir is supported by actual engineering tests.

b) Unproved Reserves are based on geological and/or engineering data similar to those used in the estimates of proved reserves, but when technical, contractual, economic or regulatory uncertainties preclude such reserves being classified as proved. They may be estimated assuming future economic conditions different from those prevailing at the time of the estimate.

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Figure 1.3: Conceptual scheme for oil and gas resources and reserves.

Unproved reserves may further be classified as probable and possible, see Fig. 1.3.

- Probable Reserves are less certain than proved reserves and can be estimated with a degree of certainty sufficient to indicate they are more likely to be recovered than not.
- Possible Reserves are less certain than proved reserves and can be estimated with a low degree of certainty, insufficient to indicate whether they are more likely to be recovered than not.

The estimation of reserves will depend upon the actual mode of petroleum recovery, which may involve either a natural-drive mechanism improved by water or gas injection, or some special technique of enhanced oil recovery (EOR).

1.7 PETROLEUM RESERVES ESTIMATION METHODS

The process of estimating oil and gas reserves for a producing field continues throughout the life of the field. There is always uncertainty in making such estimates. The level of uncertainty is affected by the following factors:

1. Reservoir type,

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2. Source of reservoir energy,
3. Quantity and quality of the geological, engineering, and geophysical data,
4. Assumptions adopted when making the estimate,
5. Available technology, and
6. Experience and knowledge of the evaluator.

Figure 1.4: Magnitude of uncertainty in reserves estimates

The magnitude of uncertainty, however, decreases with time until the economic limit is reached and the ultimate recovery is realized, see Figure 1.4.

The oil and gas reserves estimation methods can be grouped into the following categories:

1. Analogy,
2. Volumetric,
3. Decline analysis,
4. Material balance calculations for oil reservoirs,
5. Material balance calculations for gas reservoirs,
6. Reservoir simulation.

1.7.1 Analogy-Based Approach

In the early stages of development, reserves estimates are restricted to the analogy and volumetric calculations. Another producing reservoir with comparable characteristics can be used as a possible analogue for the reservoir under consideration, either by a direct well-to-well comparison or on a unit recovery basis. This can be done by determining an average oil or gas recovery per well in the analogue reservoir (e.g., 100,000 bbl/well) and applying a similar or adjusted recovery factor to the wells in the reservoir considered.

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In an analogue approach, one has to consider similarities of well spacing, reservoir rock lithofacies, rock and fluid properties, reservoir depth, pressure, temperature, pay thickness and drive mechanism. All possible differences between the analogue reservoir and the reservoir in question need to be considered to make a realistic adjustment of the recovery estimates. However, an analogue-based approach is the least accurate and little reliable method of petroleum reserve estimation, simply because perfect analogues can seldom be found.

1.7.2 Volumetric Estimates

The volumetric method, on the other hand, entails determining the areal extent of the reservoir, the rock pore volume, and the fluid content within the pore volume. This provides an estimate of the amount of hydrocarbons-in-place. The ultimate recovery, then, can be estimated by using an appropriate recovery factor. Each of the factors used in the calculation above have inherent uncertainties that, when combined, cause significant uncertainties in the reserves estimate.

The methods of reserve estimation based on reservoir data are volumetric and can be divided into deterministic and probabilistic (stochastic) estimates. The main difficulty in a volumetric estimate of resources/reserves is in the transfer of data obtained at a small scale (core analysis, lithofacies data, well logs, etc.) into a much larger scale (i.e. data "upscaling" for interwell space).

A comparison of the deterministic and probabilistic methods, however, can provide quality assurance for estimating hydrocarbon reserves; i.e. reserves are calculated both deterministically and probabilistically and the two values are compared. If the two values agree, then confidence on the calculated reserves is increased. If the two values are away different, the assumptions need to be re-examined.

1.7.2.1 Deterministic Methods

The deterministic method is by far the most common. The procedure is to select a single value for each parameter to input into an appropriate equation, to obtain a single answer.

The principle of a deterministic approach to resources/reserve estimates is to "upscale" the information derived from the wells and supported by seismic survey, into the interwell space by using an interpolation technique.

The main parameters used for a volumetric estimate in this approach are:

- The reservoir "gross" isopach map, which means the bulk thickness of the reservoir rocks (formation).

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- The reservoir "net" isopach map, which means the cumulative thickness of the permeable rock units only. The Net-to-Gross ratio (N/G) is an important parameter indicating the productive portion of the reservoir.
- The reservoir rock porosity
- The permeability and net-thickness product ($kxhN$) is important for the estimation of well production capacity
- Volume-based average saturation of water, gas and oil

Plotting these parameters as contoured maps (isopachs, isoporosity, isopermeability, etc.) provides the crucial information on their variation and distribution in the reservoir and makes it possible to evaluate the reservoir pore volume and its fractions saturated with oil and gas (hydrocarbon volume).

1.7.2.2 Stochastic Methods

The probabilistic method is more rigorous and less commonly used. This method utilizes a distribution curve for each parameter and, through the use of Monte Carlo Simulation; a distribution curve for the answer can be developed. Assuming good data, a lot of qualifying information can be derived from the resulting statistical calculations, such as the minimum and maximum values, the mean (average value), the median (middle value), the mode (most likely value), the standard deviation and the percentiles,

The probabilistic estimation of resources/reserves takes more account of the estimate uncertainty. Stochastic reservoir description is usually based on the procedure of random-number generator. This numerical technique assumes that the main reservoir properties (porosity, permeability, N/G, etc.) all have random, possibly normal, frequency distributions, with the range of values included by core and well-log data. The maximum and minimum values are specified for each of the reservoir parameters and the random number generator then "drowns data", so to speak, and then simulates their actual density distribution in the whole reservoir.

In practice, it is necessary to repeat the stochastic simulation for different "seeds" (initial boundary values) in order to assess and quantify the actual variation of a given parameter. Each numerical realisation bears an uncertainty for the reservoir characterisation, where the probabilistic rather than deterministic, is an estimate of resources/reserves.

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1.7.3 Performance Analysis

The methods of performance analysis presently used include:

- Analysis based on Material Balance Equation (MBE)
- Reservoir Simulation Models (RSM)
- Decline Curve Analysis (DCA)

The aim of all of these methods is to obtain the best reservoir performance prediction on the basis of available data. As production and pressure data from a field become available, decline analysis and material balance calculations, become the predominant methods of calculating reserves. These methods greatly reduce the uncertainty in reserves estimates; however, during early depletion, caution should be exercised in using them.

The MBE method is based on the data obtained from previous reservoir performance and PVT analysis, but involves some assumptions for the reservoir driving mechanism in order to minimise the range of possible predictions from the dataset. The method is thus adjusted differently to reservoirs containing oil, gas or oil with a gas cap (primary or secondary).

The RSM method involves a numerical simulation technique, with the matching of the production and the reservoir's previous performance (history). The discrepancy between the simulation results (prediction) and the available data is minimised by adjusting the reservoir parameters and taking into account the most likely reservoir drive mechanism (history match).

The DCA method is to predict future performance of the reservoir by matching the observed trend of the production decline with one or several standard mathematical methods of the production rate-time (hyperbolic, harmonic, exponential, etc.). If successful, such a performance analysis allows to estimate both the reserves and the future performance of the reservoir.

Decline curve relationships are empirical, and rely on uniform, lengthy production periods. It is more suited to oil wells, which are usually produced against fixed bottom-hole pressures.

In gas wells, however, wellhead back-pressures usually fluctuate, causing varying production trends and therefore, not as reliable.

The following "decline curves" from production wells are commonly used in the DCA:

- Production rate vs. time.
- Production rate vs. cumulative oil production.
- Water cut vs. cumulative oil production.
- Gas-oil ratio vs. cumulative production.
- Percentage oil production vs. cumulative oil production.
- The (p/z) ratio vs. cumulative gas production.

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Some of these decline curves are shown in Fig. 1.5.

Figure 1.5: Different ways of data representation for a decline curve analysis.

1.8 PRODUCTION PHASES

The first phase, production build up, which may exist or not depending on the drilling strategy is the increased production as wells are brought on stream. Clearly, as in some cases, wells might be predrilled through a template and then all brought on stream together when connected to production facilities, such a build up of production will, therefore, not occur.

The next stage represents the period when the productivity of the production facility is at its design capacity and the wells are throttled back to limit their productivity. This period is called the plateau phase when production is maintained at the design capacity of the facilities. Typical production rates for the plateau period cannot be presented since it depends on the techno-economics of the field.

A time will come when the reservoir is no longer able to deliver fluids to match the facilities capacity and the field goes into the decline phase. This phase can be delayed by methods to increase production. Such methods could include artificial lift, where the effort required to

lift the fluids from the reservoir is carried out by a downhole pump or by using gas lift to reduce the density of the fluid system in the well.

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lift the fluids from the reservoir is carried out by a downhole pump or by using gas lift to reduce the density of the fluid system in the well.

There comes a time when the productivity of the reservoir is no longer able to generate revenues to cover the costs of running the field, This abandonment time again is influenced by the size and nature of the operation.

Figure 1.6: Phases in production.

1.9 OIL RECOVERY TECHNIQUES

1.9.1 Primary Oil Recovery

This is the recovery obtained through the natural energy of the reservoir. The Natural Drive Mechanism types include:

1. Depletion Drive: A depletion type reservoir is a reservoir in which the hydrocarbons contained are NOT in contact with a large body of permeable water bearing sand. In a depletion type reservoir the reservoir is virtually totally enclosed by porous media and the only energy comes from the reservoir system itself.

In depletion drive reservoirs the energy comes from the expansion of the fluids in the reservoir and its associated pore space. There are two types of depletion drive reservoirs, solution gas drive reservoirs and gas cap drive reservoirs.

- Solution Gas Drive: In solution gas drive reservoirs the initial condition is where the reservoir is undersaturated, i.e. above the bubble point. Production of fluids down to the bubble point is as a result of the effective compressibility of the system. This combined compressibility (of oil, water and pores) provides the drive mechanism for depletion drive above the bubble point. Solution gas drive only occurs once the bubble point pressure has been reached.

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- Gas Cap Drive: Another kind of depletion type is where there is already free gas in the reservoir, accumulated at the top of the reservoir in the form of a gas cap. This gas cap drive reservoir, as it is termed, receives its energy from the high compressibility of the gas cap. The two significant sources of driving energy are gas cap expansion and expansion of gas coming out of solution.

2. Water Drive: reservoir is one in which the hydrocarbons are in contact with a large volume of water bearing sand.

3. Gravity drainage: or gravitational segregation is a situation where the natural density segregation of the phases is responsible for moving the fluids to the well bore.

Gravity drainage is where the relative density forces associated with the fluids cause the fluids, the oil, to drain down towards the production well.

4. Compaction Drive: Although not a common drive energy, the characteristics of its occurrence can be dramatic. Compaction drive occurs when the hydrocarbon formation is compacted as a result of the increase in the net overburden stress as the reservoir pore pressure is reduced during production.

1.9.2 Secondary Recovery

This is considered when the energy is supplemented by injection of fluids, for example gas or water, to maintain the pressure or partially maintain the pressure. The injected fluid also acts as a displacing fluid sweeping the oil to the producing wells.

1.9.3 Tertiary or Enhanced Oil Recovery (EOR)

After sweeping the reservoir with water or gas there will still be remaining oil; oil at a high saturation where the water for a range of reasons, for example; well spacing, viscosity, reservoir characteristics to name just a few, has by-passed the oil. The oil which has been contacted by the injected fluid will not be completely displaced from the porous media. Because of characteristics of the rock and the fluids a residual saturation of fluid is held

within the rock. Both of these unrecovered amounts, the by-passed oil and the residual oil are a target for enhanced recovery methods, EOR. Figure 1.7 gives a schematic representation of the various phases of development and includes the various improved recovery methods

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Figure 1.7 Oil Recovery Mechanisms.

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2.0 FUNDAMENTALS OF RESERVOIR ROCKS AND FLUID PROPERTIES

2.1 ROCK PROPERTIES

Rock properties are determined by performing laboratory analyses on cores from the reservoir to be evaluated. There are basically two main categories of core analysis tests that are performed on core samples regarding physical properties of reservoir rocks. These are Routine core analysis tests and Special tests:

Routine core analysis tests

- Porosity
- Permeability
- Saturation

Special tests

- Overburden pressure
- Capillary pressure
- Relative permeability
- Wettability
- Surface and interfacial tension

The above rock property data are essential for reservoir engineering calculations as they directly affect both the quantity and the distribution of hydrocarbons and, when combined with fluid properties, control the flow of the existing phases (i.e., gas, oil, and water) within the reservoir.

Petrophysics is the study of the physical properties of rocks. For a rock to be a reservoir rock, it must fulfill the following characteristics:

- It must have a certain storage capacity: this property is characterized by the porosity
- The fluids must be able to flow in the rock: this property is characterized by the permeability
- It must contain a sufficient quantity of hydrocarbons, with a sufficient concentration: the impregnated volume is a factor here, as well as the saturations.

The methods used to characterize reservoir rocks are essentially core analysis and well logging. Rock properties are determined by performing laboratory analyses on cores from the reservoir to be evaluated

The specifications of a reservoir rock are such that there must be a large enough capacity to store economically viable amounts of hydrocarbon and the hydrocarbon must flow at economical rates when penetrated by a well. The factors which may affect the capacity and the flow properties are the porosity, permeability, capillary pressure, compressibility and fluid saturation.

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In order to contain enough oil or gas to make production economically viable, a reservoir rock must exceed: a minimum porosity, a minimum thickness, a minimum permeability, and a minimum area.

2.1.1 Porosity

Porosity is one of the most important rock properties as a measure of the space available for accumulation of hydrocarbon fluids. The porosity of a porous medium is given the symbol of ϕ and is defined as the ratio of void space, or pore volume (V_p), to the total bulk volume (V_b) of the rock. This definition refers to an absolute porosity. This ratio is expressed either as a fraction or in percent.

$$\phi = \frac{V_p}{V_b}$$

$$\phi = \frac{V_p}{V_b} \times 100\%$$

The factors which affect the porosity (and permeability) of sandstone include the packing, size and shape of the grains, variations in size of grains, arrangement in which grains were laid down and compacted, and amount of clay and other materials which cement the sand grains together.

2.1.1.1 Absolute and effective porosity

Absolute porosity

The absolute porosity is defined as the ratio of the total pore space in the rock to that of the bulk volume. A rock may have considerable absolute porosity and yet have no conductivity to fluid for lack of pore interconnection.

Effective porosity

The effective porosity is the percentage of interconnected pore space with respect to the bulk volume. The effective porosity is the value that is used in all reservoir engineering calculations because it represents the interconnected pore space that contains the recoverable hydrocarbon fluids.

2.1.2 Permeability

The permeability of a rock is the description of the ease with which fluid can pass through the pore structure. In order to extract the fluids the rock must be permeable which requires that there be sufficiently large, interconnecting pores.

In engineering the unit of permeability is the Darcy, derived from the empirical equation known as Darcy's Law named after a French scientist who investigated the flow of water

through filter beds in 1856. His work provided the basis of the study of fluid flow through porous rock.

Darcy's law of fluid flow states that rate of flow through a given rock varies directly with the pressure applied, the area open to flow and varies inversely with the viscosity of the fluid

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flowing and the length of the porous rock. In terms of equating the parameters, the constant of proportionality in the equation is termed the permeability.

$$Q \propto \Delta p$$

$$Q \propto A$$

$$Q \propto \frac{1}{\mu}$$

where:

Q = flow rate in cm³/sec

A = cross sectional area of sample in cm²

Δp = pressure different across sample, atm

μ = viscosity in centipoise

L = length of sample in cm

k = permeability in Darcy

2.1.3 Saturation

Saturation is defined as that fraction, or percent, of the pore volume occupied by a particular fluid (oil, gas, or water). This property is expressed mathematically by the following relationship:

$$\text{Saturation} = \frac{\text{Volume of fluid}}{\text{Pore Volume}}$$

$$\text{Saturation} = \frac{V_f}{V_p}$$

$$\text{Saturation} = \frac{V_f}{V_p}$$

Applying the above mathematical concept of saturation to each reservoir fluid gives

$$\text{Water Saturation} = \frac{V_w}{V_p}$$

$$\text{Oil Saturation} = \frac{V_o}{V_p}$$

$$\text{Gas Saturation} = \frac{V_g}{V_p}$$

$$\text{Water Saturation} = \frac{V_w}{V_p}$$

$$\text{Oil Saturation} = \frac{V_o}{V_p}$$

$$\text{Gas Saturation} = \frac{V_g}{V_p}$$

$$\text{Water Saturation} = \frac{V_w}{V_p}$$

$$\text{Oil Saturation} = \frac{V_o}{V_p}$$

$$\text{Gas Saturation} = \frac{V_g}{V_p}$$

The saturation of each individual phase ranges between zero to 100%. By definition, the sum of the saturations is 100%, therefore

$$S_w + S_o + S_g = 1$$

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2.2 THERMODYNAMIC PROPERTIES OF PURE HYDROCARBON COMPOUNDS AND MIXTURES

Hydrocarbon mixtures may exist in either the gaseous or liquid state, depending on the reservoir and operating conditions to which they are subjected. To understand and predict the volumetric behaviour of oil and gas reservoirs as a function of pressure, knowledge of the physical properties of reservoir fluids must be gained.

The most important of these properties are:

- Critical pressure, p_c
- Critical temperature, T_c
- Critical volume, V_c
- Critical compressibility factor, z_c
- Acentric factor, ω
- Molecular weight, M

These fluid properties are usually determined by laboratory experiments performed on samples of actual reservoir fluids. In the absence of experimentally measured properties, it is necessary for the petroleum engineer to determine the properties from empirically derived correlations.

(a) Katz and Firoozabadi (1978) presented a generalized set of tabulated physical properties, such as average boiling point, specific gravity, and molecular weight etc, for the petroleum fractions C6 through C45.

(b) Ahmed (1985), then correlated Katz-Firoozabadi-tabulated physical properties with the number of carbon atoms of the fraction by using a regression model. The generalized equation has the following form:

$$\phi = a_1 + a_2 n + a_3 n^2 + a_4 n^3 + (a_5/n) \quad (1-1)$$

where ϕ = any physical property

n = number of carbon atoms, i.e., 6, 7, . . . , 45

$a_1 - a_5$ = coefficients of the equation and are given in Table 1-3

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(c) Riazi and Daubert (1987) developed a simple two-parameter equation for predicting the physical properties of pure compounds and undefined hydrocarbon mixtures.

The proposed generalized empirical equation is based on the use of the molecular weight M and specific gravity of the undefined petroleum fraction as the

correlating parameters. Their mathematical expression has the following form:

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(d) Edmister (1958) proposed a correlation for estimating the acentric factor of pure fluids and petroleum fractions. The equation, widely used in the petroleum industry, requires boiling point, critical temperature, and critical pressure. The proposed expression is given by the following relationship:

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Exercise

- 1.
- 2.

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(1) CORRELATIONS FOR PHYSICAL PROPERTIES OF RESERVOIR FLUIDS

There are well-established physical property correlations for the following reservoir fluids:

- Natural gases
- Crude oil systems
- Reservoir water systems

Natural gases

Generally, the natural gas is a mixture of hydrocarbon and non-hydrocarbon gases. The hydrocarbon gases that are normally found in a natural gas are methanes, ethanes, propanes, butanes, pentanes, and small amounts of hexanes and heavier. The non-hydrocarbon gases (i.e., impurities) include carbon dioxide, hydrogen sulfide, and nitrogen.

Knowledge of pressure-volume-temperature (PVT) relationships and other physical and chemical properties of gases is essential for solving problems in natural gas reservoir engineering. These properties include:

- Apparent molecular weight, M_a
- Specific gravity, γ_g
- Compressibility factor, z
- Density, ρ_g
- Specific volume, v
- Isothermal gas compressibility coefficient, c_g
- Gas formation volume factor, B_g
- Gas expansion factor, E_g
- Viscosity, μ_g

The above gas properties may be obtained from direct laboratory measurements or by prediction from generalized mathematical expressions.

(i) If we assume an ideal situation, then the PVT relationships is given by:

Ideal gas law equation

$$pV = nRT \quad (2 - 1)$$

where p = absolute pressure, psia

V = volume, ft³

T = absolute temperature, °R

n = number of moles of gas, lb-mole

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R = the universal gas constant which, for the above units, has the

value 10.730 psia ft³/lb-mole °R

The density of the gas can be given by:

where ρ_g = density of the gas, lb/ft³

(ii) Assuming non-ideal situation; since natural gas is not a pure components but a mixture of hydrocarbon components, the overall physical and chemical properties can be determined from the physical properties of the individual components in the mixture by using appropriate mixing rules.

The basic properties of gases are commonly expressed in terms of the apparent molecular weight, standard volume, density, specific volume, and specific gravity.

These properties are defined as follows:

Apparent Molecular Weight

Standard Volume

The standard volume is then defined as the volume of gas occupied by 1 lb-mol of gas at standard conditions of 14.7 psia and 60°F.

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Density

Specific Volume

The specific volume is defined as the volume occupied by a unit mass of the gas.

Specific Gravity

The specific gravity is defined as the ratio of the gas density to that of the air; with both densities measured at the standard pressure P_{sc} and standard temperature T_{sc}

or

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Example

1. Three pounds of n-butane are placed in a vessel at 120°F and 60 psia. Calculate the volume and density of the gas assuming ideal gas behaviour. Given that the molecular weight of n-butane is, $M = 58.123$

Solution

2. A gas well is producing gas with a specific gravity of 0.65 at a rate of 1.1MMscf/day. The average reservoir pressure and temperature are 1,500 psi and 150°F. Calculate:

- Apparent molecular weight of the gas
- Gas density at reservoir conditions
- Flow rate in lb/day

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Solution

-
-
-

3. A gas well is producing a natural gas with the following composition:

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Assuming ideal gas behaviour, calculate:

- Apparent molecular weight
- Specific gravity
- Gas density at 2000 psia and 150°F
- Specific volume at 2000 psia and 150°F

Solution

- Applying
-
-
-

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Real Gas Situation

At high pressure and high temperature, the PVT relationship is more accurately represented by:

where the gas compressibility factor z is a dimensionless

quantity and is defined as the ratio of the actual volume of n-moles of gas at T and p to the ideal volume of the same number of moles at the same T and p:

Studies of the gas compressibility factors for natural gases of various compositions have shown that compressibility factors can be generalized with sufficient accuracies for most engineering purposes when they are expressed in terms of the following two dimensionless properties:

- Pseudo-reduced pressure
- Pseudo-reduced temperature

$$Z = f(P_{pr}, T_{pr})$$

These dimensionless terms are defined by the following expressions:

where

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P = system pressure, psia

P_{pr} = pseudo-reduced pressure, dimensionless

T = system temperature, °R

T_{pr} = pseudo-reduced temperature, dimensionless

P_{pc}, T_{pc} = pseudo-critical pressure and temperature, respectively, and defined by the following relationships:

Standing and Katz (1942) presented a generalized gas compressibility factor chart as shown in Figure 2-1. The chart represents compressibility factors of sweet natural gas as a function of P_{pr} and T_{pr}. This chart is generally reliable for natural gas with minor amount of non-hydrocarbons. It is one of the most widely accepted correlations in the oil and gas industry.

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Example 1

A gas reservoir has the following gas composition: the initial reservoir pressure and temperature are 3000psia and 180°F, respectively.

Calculate the gas compressibility factor under initial reservoir conditions.

Solution

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Determine the z-factor from Figure 2-1, to give:

$$z = 0.85$$

Example 2

Using the data below and assuming real gas behaviour, calculate the density of the gas phase under initial reservoir conditions.

Compare the results with that of ideal gas behaviour.

Solution

From the chart

Given that

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Calculate the density of the gas assuming an ideal gas behaviour ($Z = 1$)

The results of the above example show that the ideal gas equation estimated the gas density with an absolute error of 15% when compared with the density value as predicted with the real gas equation.

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PSEUDO-CRITICAL PRESSURE AND PSEUDO-CRITICAL TEMPERATURE OF GASES WHERE COMPOSITION OF A NATURAL GAS IS NOT AVAILABLE

Standing (1977) provided a mathematical expression of the graphical correlation by Brown et al. (1948) for a convenient approximation of the pseudo-critical pressure and pseudo-critical temperature of gases when only the specific gravity of the gas is available.

Example 3

Using the data below and assuming unavailability of pseudo-critical temperature and pseudo-critical pressure data,

calculate the density of the gas phase under initial reservoir conditions.

Given that for a natural gas system

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Solution

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Absolute and Effective Porosity

Absolute porosity

Effective porosity

Application of the effective porosity

One important application of the effective porosity is its use in determining the original hydrocarbon volume in place. Consider a reservoir with an areal extent of A acres and an average thickness of h feet. The total bulk volume of the reservoir can be determined from the following expressions:

The reservoir pore volume PV can then be determined by combining Equations 4-4 and 4-5 with 4-3.

NB: 1 Acre = 43,560 square feet and 1 barrel = 5.614583 cubic feet

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Example 1

An oil reservoir exists at its bubble-point pressure of 3000 psia and temperature of 160°F. The oil has an API gravity of 42° and gas-oil ratio of 600 scf/STB. The specific gravity of the solution gas is 0.65. The following additional data are also available:

- Reservoir area = 640 acres
- Average thickness = 10 ft
- Connate water saturation = 0.25
- Effective porosity = 15%

Calculate the

(a) Oil Initially In Place (OIIP) in bbl

(b) Stock Tank Oil Initially In Place (STOIIP) in STB.

Solution

(a) Determine the specific gravity of the stock-tank oil from

Calculate the initial oil formation volume factor (FVF) by applying Standing's equation,

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Calculate the pore volume from

$$\text{Pore volume} = 7758 (640) (10) (0.15) = 7,447,680 \text{ bbl}$$

$$\text{OIIP} = \text{Pore Volume} (1 - S_{wc})$$

$$\text{OIIP} = 7,447,680 (1 - 0.25) = 5,585,760 \text{ bbl}$$

(b) Calculate the Stock Tank Oil Initially In Place (STOIIP).

$$\text{STOIIP} = \text{Pore Volume} (1 - S_{wc})$$

Boi

$$\text{Initial oil in place} = 7,447,680 (1 - 0.25)/1.306 = 4,276,998 \text{ STB}$$

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2.2 COMPOSITIONAL DESCRIPTION FOR RESERVOIR ENGINEERING

Petroleum engineers require a compositional description tool to use as a basis for predicting reservoir and well fluid behaviour. The two approaches that are commonly used are the multicomponent compositional model and the two component black oil model. The latter simplistic approach has been used for many years to describe the composition and behaviour of reservoir fluids. It is called the "Black Oil Model".

2.2.1 Black Oil Model

Two component description of the reservoir fluid consisting of gas dissolved in oil (solution gas) and stock tank oil. Compositional changes with varying pressure and temperature are ignored. To those appreciating thermodynamics this simplistic two component model is difficult to cope with.

Figure 2.1: "Black Oil Model"

Terms such as 'Gas Oil Ratio' and 'Formation Volume Factor' are black oil model terms. Associated with the black oil model are Black Oil model definitions in relation to Gas Solubility and Formation Volume Factors.

2.2.2 Compositional Model

The compositional model is based on the paraffin series C_nH_{2n+2} . To keep the number of components in the model manageable, long chain members are grouped together and given an average property. These compounds are termed collectively as the 'C+ fraction'. Typically

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this covers the hydrocarbons above Heptane and therefore is called the C7+ fraction, which is characterised using the terms Apparent Molecular Weight and Specific Gravity.

Figure 2.2: Compositional Model

2.2.3 Comparison of Reservoir Fluid Models

It is useful to summarise the differences between the Black Oil Model approach compared to the Compositional Model in predicted fluid properties.

The suitability of the two approaches is largely related to the nature of the fluid. For heavier oils where there are low GOR's as compared to volatile oils with high GOR's, black oil models are likely to be suitable. For the more volatile systems where there are more significant compositional variations in a reservoir as pressure is depleted, compositional models are considered more capable of predicting fluid behaviour. The computational requirements of compositional models used to be a restriction when carrying out large reservoir simulation. The continued development of computing and associated equations of state modelling reduces these former restrictions.

The list below gives a summary comparison of the two approaches.

Black Oil Models

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- Components - solution gas and stock tank oil
- Bo, Rg, etc.
- Empirical correlations

Compositional Models

- N components based on paraffin series
- Equation of state based calculations

2.3 PVT FLUID PROPERTIES

Inspection and rationalisation of the pressure-volume-temperature (PVT) fluid properties must be the opening move in the study of any oilfield since the PVT functions, which relate surface to reservoir volumes, are required in practically every aspect of reservoir engineering: calculation of hydrocarbons in place, pressure depth regimes, any form of recovery calculations and to assure correct design of surface facilities.

2.3.1 BASIC PVT PARAMETERS

For an oil reservoir these are:

- BO: oil formation volume factor (FVF), which is the number of reservoir barrels of oil and dissolved gas that must be produced to obtain one stock tank barrel of stable oil at the surface (rb/stb).
- RS: solution or dissolved gas/oil ratio (GOR), which is the number of standard cubic feet of gas produced with each stock tank barrel of oil that was dissolved in the oil in the reservoir (scf/stb).
- Bg: gas formation volume factor, which is the volume in barrels that one standard cubic foot of gas at the surface occupies as free gas in the reservoir (rb/scf).

All three are strictly functions of pressure at the (assumed) constant reservoir temperature.

The first two allow for the fact that oil in the reservoir, at high pressure and temperature, contains dissolved gas that is released as it is produced to the surface. For example, if $BO = 1.45$ rb/stb and $RS = 500$ scf/stb, then 1.45 rb of oil and dissolved gas must be produced to obtain 1 stb of oil, releasing 500 scf of gas at the surface. As mentioned above, all three functions are required to relate reservoir volumes to those measured at the surface. The engineer works physically in terms of reservoir volumes yet must use practical surface measured volumes in equations. Thus, in setting-up the basic equation of waterdrive in terms of: injection rate = gross production rate (at constant pressure), it becomes:

$$Q_{inj} = Q_o + Q_g \left(\frac{BO}{B_o} \right)$$

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in which all the rates, q , are surface measured values (stb/d) but multiplication by the FVFs (rb/stb), establishes the equation as an "underground" balance (rb/d), in which Q_o is the flow rate of the single phase oil containing R , (scf/stb) of dissolved gas while Q_g is the reservoir flow rate of the produced water which may contain a small amount of dissolved gas, hence the need for the water FVF B_w , which is usually close to unity.

All three PVT parameters are measured in laboratory experiments conducted at the constant reservoir temperature. The results are then dependent on the declining pressure alone.

Fig. 2.3. PVT functions for a moderate volatility oil. (a) FVF; (b) solution GOR.

Typical shapes of the BO and RS functions for a moderate volatility oil are shown in Fig. 2.3. At the initial pressure, p_i , the oil is described as "undersaturated", meaning that if there was free gas available it would be dissolved - but there is none. The oil is therefore in the single phase state containing dissolved gas. Eventually, through continued depletion, the pressure drops to the bubble point, p_b , at which the oil becomes gas saturated. Below this pressure gas is liberated from the oil giving rise to the presence of two phases: oil + dissolved gas and free gas.

Above the bubble point, as the pressure declines, the oil must expand hence BO increases. Since the oil compressibility is usually fairly small, however, the extent of the expansion is slight. Also, above the bubble point each stb of oil produced contains the constant R_{si} scf of dissolved gas, since none has been liberated in the reservoir.

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Figure 2.4: Production of reservoir hydrocarbons (a) above bubble point pressure, (b) below bubble point pressure

Below the bubble point, matters are more complicated. Gas is freed from the oil in situ, consequently B , decreases continuously since the dissolved gas content is reduced. The decrease in oil volume from its maximum value at the bubble point is referred to as the "shrinkage", $BO_b - BO$, at which point the amount of gas released is $R_{Si} - RS$ scf/stb. Producing below the bubble point is invariably "messy", because the gas viscosity is typically fifty times lower than that of oil, therefore, according to Darcy's law, it is capable of travelling much

faster. This leads to recovery under the so-called "solution gas drive" condition. It would be wrong to state that production below the bubble point necessarily results in a low oil recovery but on most occasions it does. There is an uncontrollable element associated with the process due to the completely different mobilities of the oil and free gas, which leads to the tendency to strip the gas from the reservoir prematurely at the expense of the less mobile oil

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Figure 2.5: Application of PVT parameters to relate surface to reservoir hydrocarbon volumes; above bubble point pressure.

Figure 2.6: Application of PVT parameters to relate surface to reservoir hydrocarbon volumes; below bubble point pressure

In volatile oil reservoirs, in which the oil contains a large volume of dissolved gas that will be released at the surface (typically: $BO_i = 2.8$ rb/stb, $RS_i = 3000$ scf/stb) the variation in FVF with pressure is as shown in Fig. 2.7. Even above the bubble point, the high dissolved gas content

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in the oil means that the change in volume with pressure is very significant. This is reflected in the oil compressibility which might typically have a value of $60-80 \times 10^{-6}$ /psi. Problems start, however, immediately below the bubble point where the shrinkage, $BO_b - BO_i$, is very dramatic and, as the oil shrinks in situ, it gives the gas greater freedom to move through the formation and be produced in excess quantities. The situation can be so extreme that what was accepted as an oil field one day can appear to behave like a gas field the next, simply by passing through the bubble point pressure. This is reflected in the highly unfavourable oil relative permeability which inhibits its flow. Therefore, in such volatile oil reservoirs it is obviously preferable to aim at operating in the undersaturated pressure range just above the bubble point (to take advantage of the natural oil expansion) through the injection of either water or gas to provide pressure maintenance.

Fig. 2.7. FVF of a volatile oil.

The third PVT parameter, the gas FVF, B_g (rb/scf), is only required in initially undersaturated reservoirs once gas has been liberated below the bubble point pressure. It then relates free gas volumes in the reservoir (rb) to the surface volume (scf). Alternatively, if there is a gas cap at initial conditions, B_g would be required from the start. On account of its units (rb/scf) the FVF has a very small value at high pressure, typically, 0.0007 rb/scf. While this should not be of concern to us these days, as a historical note, it is worth mentioning the problems it posed our predecessors in the subject: armed computationally with only slide rules, with which the difficulty was trying to figure out where the decimal point should be located! The B_g function has a rather unfortunate, near hyperbolic shape (Fig. 2.8), causing some difficulties in interpolation/extrapolation. When working in the low-pressure range, the engineer should be careful to choose very small pressure increments when entering the B_g

function to computer programs to assure accuracy in its use.

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Fig. 2.8. Gas FVF as a function of pressure

2.4. COLLECTION OF FLUID SAMPLES (SAMPLING RESERVOIR FLUIDS)

Samples of the reservoir fluid are usually collected at an early stage in the reservoir's producing life and dispatched to a laboratory for the full PVT analysis. There are basically two ways of collecting such samples, either by direct subsurface sampling or by surface recombination of the oil and gas phases. Whichever technique is used the same basic problem exists, and that is, to ensure that the proportion of gas to oil in the composite sample is the same as that existing in the reservoir. Thus, sampling a reservoir under initial conditions, each stock tank barrel of oil in the sample should be combined with Rsi standard cubic feet of gas i.e. 1 stb Oil + RSi scf Gas

2.4.1 Subsurface Sampling

In this case a subsurface sampler is lowered into the well and kept opposite the producing layer for a sufficiently long time, Figure 1. Many types of bottom hole samplers have been devised and described in the literature.

Subsurface samples can only be representative of the reservoir contents when the pressure at the point of sampling is above or equal to the saturation pressure. If this condition is not fulfilled, one should take a surface sample. Even at pressures close to the saturation pressure there is a serious possibility of sample integrity being lost as a result of the system going two phase during transfer to the sample chamber.

A simple test can be carried out in the field at the well site to find out whether a reliable sample has been obtained. In this test the sample cylinder is pressurised either by using a piston cell or using mercury as the displacing fluid. Whereas mercury was the most common fluid to be used as a pressure transfer and volume change fluid, because of toxic and other

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concerns its use is diminishing. From the relation between injection pressure and volume of mercury injected, the following properties are derived:

- (1) the pressure existing within the sampler when it is received at the surface;
- (2) the compressibility of the material within the sampler;
- (3) the bubble-point pressure of the contents of the sampler.

If two or three samples taken at short time intervals show the same measured properties, it is highly probable that good samples have been obtained.

In recent years there have been considerable advances on downhole fluid sampling.

The contents of the sampler are transferred by means of a mercury pump into a high pressure shipping container. Two fillings of a sampler of 600 ml capacity are usually sufficient for a complete PVT analysis. In recent years mercury has been replaced using the

application of piston cells by some companies.

Subsurface sampling is generally not recommended for gas-condensate reservoirs nor for oil reservoirs containing substantial quantities of water.

Figure 2.9: Subsurface Sampling

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Fig. 2.10: Subsurface collection of PVT sample

2.4.2 Surface recombination sampling

In collecting fluid samples at the surface, separate volumes of oil and gas are taken at separator conditions and recombined to give a composite fluid sample. The surface equipment is shown schematically in fig. 2.11.

Fig. 2.11: Collection of a PVT sample by surface recombination

The well is produced at a steady rate for a period of several hours and the gas oil ratio is measured in scf of separator gas per stock tank barrel of oil. If this ratio is steady during the

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period of measurement then one can feel confident that recombining the oil and gas in the same ratio will yield a representative composite sample of the reservoir fluid. In fact, a slight adjustment must be made to determine the actual ratio in which the samples should be recombined. This is because, as shown in fig. 2.11, the oil sample is collected at separator pressure and temperature whereas the gas oil ratio is measured relative to the stock tank barrel, thus the required recombination ratio is

Dimensionally, the measured gas oil ratio must be multiplied by the shrinkage factor from separator to stock tank conditions. This factor is usually determined in the laboratory as the first stage of a PVT analysis of a surface recombination sample by placing a small volume of the oil sample in a cell at the appropriate separator conditions and discharging it (flash expansion) to a second cell maintained at the field stock tank conditions. During this process some gas will be liberated from the separator sample, due to the reduction in pressure and temperature, and the diminished stock tank oil volume is measured, thus allowing the direct calculation of S . In order to be able to perform such an experiment it is important that the engineer should accurately measure the pressure and temperature prevailing at both separator and stock tank during sampling and provide the laboratory with these data.

One of the attractive features of surface recombination sampling is that statistically it gives a reliable value of the producing gas oil ratio measured over a period of hours; furthermore, it enables the collection of large fluid samples. Of course, just as for subsurface sampling, the surface recombination method will only provide the correct gas oil ratio if the pressure in the vicinity of the well is at or above bubble point pressure. If not, the surface gas oil ratio will be too low or too high, depending upon whether the free gas saturation in the reservoir is below or above the critical saturation at which gas will start to flow. In this respect it

should be emphasized that PVT samples should be taken as early as possible in the producing life of the field to facilitate the collection of samples in which the oil and gas are combined in the correct ratio.

2.5 DETERMINATION OF THE BASIC PVT PARAMETERS IN THE LABORATORY AND CONVERSION FOR FIELD OPERATING CONDITIONS

2.5.1 PVT APPARATUS

The apparatus required for PVT analysis consists of:

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- (a) apparatus for the transfer and the recombination of separator oil and gas samples;
- (b) apparatus for measuring gas-volumes and for performing separator tests;
- (c) the PV cell and displacement pumps and dispensing cell;
- (d) high pressure viscometer;
- (e) gas chromatograph.

Diagrams of the layouts for different systems are given in Figures 2.12.

Figure 2.12: Equipment for the Study of Subsurface Samples

2.5.2 RECOMBINATION OF THE SURFACE OIL AND GAS SAMPLES

The GOR given by the field refers to the separator tank gas-oil ratio. In order to recombine the separator gas and the separator oil in the correct ratio, the volumetric ratio between tank oil and separator oil is determined in the laboratory 85% of the cylinder containing separator oil is occupied by oil. There is a gas cap on top. This precaution is taken in view of the long transport time and the risk of great fluctuations in temperature involved.

After the gas cap has been dissolved by pressing water into the cylinder at a pressure higher than the separator pressure a given quantity of oil is then flashed at pressure and room temperature through a separator operating under the same conditions as the tank in the field i.e. at the same pressure and temperature. The collected tank oil is weighed and the

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density determined, after which the volume of oil is known. The volume and the density of the liberated gas (tank gas) are also determined. From the above measurements, the tank gas/tank oil ratio is also known. If the volume of tank oil is lower than the corresponding quantity of separator oil we speak of shrinkage, in the opposite case of expansion; shrinkage occurs when a large quantity of gas is produced and expansion occurs when a small quantity of gas is dissolved in the separator oil. When the shrinkage or the expansion is known, recombination can take place on a tank-oil basis.

2.5.3 PVT TESTS

At this stage we should remind ourselves of the main applications of the PVT data.

The three main application areas are;

- to provide data for reservoir calculations,
- to provide physical property data for well flow calculations
- for surface facility design.

Although all are cited as users of the data, the reservoir calculation requirement has provided the main driving force for the tests to be carried out. In surface facility design for example the more simplistic black oil approach around which the PVT analysis is structured is considered too limiting, and the main data for this application is the compositional analysis of the fluids.

Over recent years, as the data is subsequently applied to computer based simulation tools, the ability to handle more complex descriptions of the fluids has led to more extended compositional analysis, beyond the C7+ limit which was the basis for many years. It is common practice in some PVT laboratories to measure co-position to C28 and then define a C29+ component

In reservoir calculations the PVT tests and subsequent report provides the source of the reservoir engineering properties necessary to describe the behaviour of the reservoir over its development and production. The tests conducted therefore have to take into consideration the processes going on both above and below the saturation pressure.

There are four main PVT tests for oil systems plus associated compositional analysis:

1. Flash vaporisation or relative volume tests: Constant composition expansion of the undersaturated oil from initial pressure to the bubble point.
2. Differential test: Differential vaporization: expansion below the bubble point with varying fluid composition.
3. Separator tests: Separator flash expansion experiments.

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4. Viscosity measurements: Measurement of oil and gas viscosities as functions of pressure.
5. Compositional measurements: Compositional analysis of the reservoir fluid.

All these experiments are performed under isothermal conditions at the reservoir temperature.

A simple layout of a PVT facility and PV cell with its associated equipment is given in figure 2.13 and 2.14 respectively.

Figure 2.13: Simple Schematic of PVT Facility for Oil Tests

Fig. 2.14: Schematic of PV cell and associated equipment

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2.5.3.1: Flash Expansion Experiment

A volume of undersaturated oil is expanded in a PV-cell in which all of the fluid is contained within the cell during a series of pressure decrements: usually from the initial pressure (or

above) to the bubble point. After each pressure step, the oil volume is measured and reported as a relative volume (RV):

□

□□ = (□□/□□□)

□□

in which V_b is the volume at the bubble point. The RV is necessarily less than unity and decreases with increasing pressure. Oil density and compressibility are also reported. The bubble point pressure is manifest, as described above, by the sudden large change in volume for a small change in pressure, due to the evolution of high compressibility gas.

2.5.3.2: Differential Liberation Experiment

The differential vaporization experiment defines the changing FVF and solution GOR below the bubble point pressure. Gas saturated oil is charged to a PV-cell and the pressure is reduced in steps below the bubble point; after each, the evolved gas is removed from the cell and its volume measured along with the volume of gas saturated oil remaining in the cell. Once the pressure has been reduced to atmospheric, the temperature is decreased from the reservoir value to 60°F meaning that the final state is the stock tank condition (14.7 psia, 60°F) at which the remaining oil in the cell is referred to as the "residual" volume (stb), to which all oil volumes are related at any stage of depletion.

From the volumetric measurements of the oil and gas it is possible to calculate:

B_{od} = FVF of the oil at each stage of depletion (rb/stb).

R_{sd} = corresponding solution GOR (scf/stb).

The subscript "d" relates to the differential experiment and the initial values of these parameters at the bubble point are B_{obd} (rb/stb) and R_{sid} (scf/stb) respectively.

NOTE:

The main difference between the two types of experiment shown in fig. 2.15 (a) and (b) is that in the flash expansion no gas is removed from the PV cell but instead remains in equilibrium with the oil. As a result, the overall hydrocarbon composition in the cell remains unchanged. In the differential liberation experiment, however, at each stage of depletion the liberated gas is physically removed from contact with the oil and therefore, there is a continual compositional change in the PV cell, the remaining hydrocarbons becoming

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progressively richer in the heavier components, and the average molecular weight thus increasing.

Fig. 2.15: Illustrating the difference between (a) flash expansion, and (b) differential liberation

2.5.3.3: Separator Tests

The object of these tests are to examine the influence of separator pressure and temperature on formation volume factor, gas/oil ratio, gas density and tank-oil density.

These tests are not driven by those who will be responsible for the optimised separation process if the field ultimately is developed. They are carried out to give an indication of the oil shrinkage and GOR which occurs when the fluids are produced to surface conditions. It should be emphasised at this stage that there is not a unique value for the formation volume factor and solution gas-oil ratio. It depends on the stages and conditions of separation through which the fluids pass.

With the equipment available, a single test or a multiple separation test can be carried out. The systems is set up as shown in the schematic below, Figure 2.16.

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Figure 2.16: Schematic of a Two Stage Separator Test

The procedure for the separation test is as follows. The starting point is oil in the PVT cell at its reservoir bubble point, i.e. the same starting condition as the differential test. Fluid is displaced from the PVT cell ensuring that the PVT cell contents remain at bubble point pressure. The gas and liquids are collected from the separation stage(s) and their respective properties measured. The final stage is at stock tank conditions. The resulting fluid is termed stock tank oil.

A single separator test is carried out by flashing reservoir liquid at bubble point pressure and reservoir temperature through the separator operating at the average annual temperature and at pressures which may be expected in the field. The difference in results when using a single or a double separator is that in the former case the total gas/oil ratio is higher, the shrinkage is greater and the density of the tank oil is higher than in the latter case.

The main objectives of the separator test are in combination with the flash vaporisation and differential tests to provide formation volume factor and solution gas-oil ratios over a full pressure range above and below the bubble point. In quoting these values it is important to recognise that the values are separator condition specific.

2.5.3.4: Viscosity

The viscosity is measured at reservoir temperature and at different pressures both above and below saturation pressure. It is important for viscosity measurements below the bubble point to generate the fluid for study by a differential mode to simulate the nature of the fluid that would exist at these conditions.

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Viscosity measurements were largely carried out with a “rolling ball” high pressure viscometer consisting of a highly polished-steel capillary of 1/4" dia. which can be closed at the top by means of a plunger and is provided at the bottom with a contact which is connected with an amplifier. A steel or platinum ball rolls in the capillary, its diameter hence slightly smaller than that of the capillary. When the ball reaches the bottom, it makes contact between the wall of the capillary and the point of contact, as a result of which a circuit is closed and a whistling sound is heard. The time of rolling is a measure of the

viscosity.

In recent years the pressure drop along a capillary tube of known length and internal dimensions has been used. The viscosity being calculated using the Poiseuille equation, the laminar flow pressure drop equation for a pipe of a particular diameter and length.

Although being used it is clearly restricted by operating under a fixed flow regime, laminar and velocities to ensure that there is a sufficient pressure drop to measure and not too large to influence physical properties.

2.5.3.5: Hydrocarbon Analysis

A hydrocarbon analysis is made of the methane to an upper paraffin fraction of the recombined surface (or subsurface) sample. Historically this upper limit was C6 and the remainder lumped as a C7+ fraction. C10. Higher C numbers are now used as analytical methods enable even higher levels of characterisation. The plus component having been separated in a distillation column needs to be characterised by its specific gravity and its apparent molecular weight. The latter is achieved using a depression of freezing point method. The different components are determined by means of gas/liquid chromatography. If the description is based solely on paraffins then non paraffinic components are added to the next higher paraffin.

The value of a higher characterisation is particularly helpful to process engineering considerations, say up to C30 because at lower temperatures long chain hydrocarbons will form a solid phase (such as wax) and adhere to surfaces. At reservoir conditions they are in the liquid phase and therefore do not affect the reservoir flow process.

In some cases the fluids produced in the final separation stage are identified to a higher C number together with aromatic and naphtha components. Compositions are also made of the produced gases from the various tests.

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2.5.4 COMPLETE PVT ANALYSIS

The complete PVT analysis for oil, provided by most laboratories, usually consists of the following experiments and calculations.

a) Compositional analysis of the separator oil and gas, for samples collected at the surface, together with physical recombination or compositional analysis of the reservoir fluid collected in a subsurface sample.

Such analyses usually give the mole fractions of each component up to the hexanes. The hexanes and heavier components are grouped together, and the average molecular weight and density of the latter are determined.

b) Flash expansion conducted at reservoir temperature. This experiment determines

- the bubble point pressure
- the compressibility of the undersaturated oil
- the total volume v_t of the oil and gas at each stage of depletion.

c) Differential liberation experiment to determine

- E , Z , F and v_o , with F and v_o measured relative to the unit volume of bubble point oil.

Alternatively, by measuring $c_b d$ during the last stage of the differential liberation, the above data can be presented as

- E , Z , $R_{si d} - R_{s d}$ (or just $R_{s d}$) and $B_o d$, with $R_{s d}$ and $B_o d$ measured relative to residual oil volume. In addition, the gas gravity is measured at each stage of depletion.

d) Measurement of the oil viscosity at reservoir temperature (generally using the rolling ball viscometer), over the entire range of pressure steps from above bubble point to atmospheric pressure. Gas viscosities are normally calculated at reservoir temperature, from a knowledge of the gas gravity, using standard correlations.

e) Separator tests to determine the shrinkage, $c_b f$, and solution gas oil ratio, $R_{si f}$, of unit volume of bubble point oil (1 barrel) when flashed through various separator combinations. Instead of actually performing these tests, in many cases the results are obtained using the phase equilibrium calculation technique.

f) Composition and gravity of the separator gas in the above separator tests.

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2.5.5 COMPARISON OF LABORATORY AND FIELD PVT DATA

No matter how careful the engineer may be in collecting fluid samples for laboratory PVT analysis, it seldom happens that the resulting functions match exactly the P W relations demonstrated by the field itself, once it has been brought on continuous production. There can be many reasons for the disparities, such as deferring the sampling until too late in the test or possibly because the separators installed are not quite as theoretically planned.

But perhaps the main reason is simply that if a well is sampled at the appraisal stage during the course of a DST, it must be appreciated that it is producing in a non-equilibrium state, especially with respect to conditions in the vicinity of the wellbore and the collection of completely representative samples in a few "small bottles" is too much to be expected.

Although laboratory techniques are very sophisticated, the experimental PVT results are only as reliable as the fluid samples collected and should never be regarded as more than a good approximation to the real fluid properties. Laboratory PVT results must be calibrated and, as is often necessary, corrected to match field observations after the start of production. At this time, the engineer should monitor the production data very carefully and, in particular, the producing GOR through the separator configuration used in the field and seek any evidence that the reservoir has passed through the bubble point pressure, which will occur in depletion type fields.

The typical GOR behaviour of fractured and homogeneous acting reservoirs is depicted in Fig. 2.7a. In the former, there is usually little or no development of a critical gas saturation, S_{gc} , below the bubble point, which must be exceeded before the liberated gas becomes

mobile. Instead the free gas tends to flow towards the wellbore through the fracture system as soon as it is released. In this case the saturation pressure can be directly related to the prevailing reservoir pressure at which the GOR begins to rise sharply above the value of, R_{Si} defined during the initial phase of production at undersaturated conditions (Fig. 2.7a).

Fig. 2.7. (a) GOR behaviour of fractured and homogeneous reservoirs. (b) Correction of laboratory measured GOR for field conditions.

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In the case of a more homogeneous reservoir, the development of a critical gas saturation is usually observed below the bubble point and is manifest as a slight decrease in the producing GOR as the small volume of gas is trapped in the reservoir rather than produced to the surface (Fig. 2.7a). The bubble point can then be defined as the pressure at which the decline in GOR first occurs, although this is often a more subtle observation than for fractured reservoirs. Other ways of confirming that the reservoir pressure has fallen below the bubble point are by observation of:

- an increase in the API gravity of the oil
- noting any sudden reduction in the productivity of wells

If it is determined, by direct observation of reservoir performance, that the initial GOR is different from that determined in the PVT analysis, then it becomes necessary to correct the laboratory PVT functions. If, as shown in Fig. 2.7b, the field GOR exceeds that determined by experiment and the elevated bubble point pressure has also been observed, then the RS trend below the bubble point can simply be "eye-balled" in the knowledge that at atmospheric pressure $p \rightarrow 0$. A more scientific approach might be to use standard correlations to predict the decline or even more sophisticated EOS thermodynamic software packages. If the bubble point has not been directly observed the latter techniques must be applied to determine it and, whether the bubble point has been observed or not, they need to be used to correct the oil FVF both above and below the bubble point pressure.

The PVT properties of reservoir fluids can be somewhat more complex than determined in the laboratory by performing experiments on samples collected at the appraisal stage of field development and can vary throughout the lifetime of a project. Consequently, the reservoir engineer should take advice from specialised process engineers before finalising the input to studies, whether analytical or using numerical simulation.

Comparison of laboratory and field parameters for the differential liberation experiment is shown in the table below:

Table 2.1: Laboratory and field parameters of differential liberation experiment

Laboratory Differential Field Parameter Conversion

Parameter

R_{Si} R_{Si} R_{Si}

$R_{Si} = (R_{Si}/R_{Si})$

R_{Si}

F =

$$F = \frac{V_{oi}}{V_{oi} - 5.615 \left(\frac{V_{gi}}{V_{oi}} \right)}$$

where

E = 1

$$F = \left(\frac{V_{oi}}{V_{oi}} \right)$$

5.615

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2.6 CALCULATION OF HYDROCARBON VOLUMES

The volume of oil and gas in-place depends on a number of parameters :

- The aerial coverage of the reservoir, A
- The thickness of the reservoir rock contributing to the hydrocarbon volume. h
- The pore volume, as expressed by the porosity, ϕ , the reservoir quality rock.
- The proportion of pore space occupied by the hydrocarbon (the saturation), 1-Sw

Oil Initially In Place (OIIP)

Consider a reservoir which is initially filled with liquid oil. The oil volume in the reservoir (oil in place) is

$$V_{oi} = A \phi (1 - S_{wi}) = A h \phi (1 - S_{wi})$$

Where:

A= average area

h = net thickness. Net thickness = gross thickness x net: gross ratio

V = the net bulk volume of the reservoir rock

ϕ = the porosity, or volume fraction of the rock which is porous, and

S_{wi} = the connate or irreducible water saturation and is expressed as a fraction of the pore volume.

The product $V\phi$ is called the pore volume (PV) and is the total volume in the reservoir which can be occupied by fluids. Similarly, the product $V\phi (1-S_{wc})$ is called the hydrocarbon pore volume (HCPV) and is the total reservoir volume which can be filled with hydrocarbons either oil, gas or both.

Stock Tank Oil Initially In Place (STOIIP)

To convert volumes at reservoir conditions to stock tank conditions, formation volume factors are required where BO are the oil formation volume factor. The expression of original oil in place is termed the STOIIP. When expressed as stock tank volumes, equation above is divided by the formation volume factor

$$A \phi (1 - S_{wi})$$

$$STOIIP = \left(\frac{V_{oi}}{B_o} \right)$$

where

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Recoverable reserves

The recovery factor, RF, indicates the proportion of the in-place hydrocarbons expected to be recovered. To convert in place volumes to reserves we need to multiply the STOIP by the recovery factor. The equation describing this method of applying equity is:

$$\text{Reserves} = \text{STOIP} \times \text{RF}$$

$$\text{Reserves} = \text{STOIP} \times \text{RF}$$

$$\text{Reserves} = \text{STOIP} \times \text{RF}$$

$$\text{Reserves} = \text{STOIP} \times \text{RF}$$

Movable oil

This is defined as the oil that can be physically involved during a secondary recovery flood by either water or gas injection. The equity equation can be expressed as:

$$\text{Movable Oil} = \text{STOIP} \times (\text{RF} - \text{Residual Saturation})$$

$$\text{Movable Oil} = \text{STOIP} \times (\text{RF} - \text{Residual Saturation})$$

$$\text{Movable Oil} = \text{STOIP} \times (\text{RF} - \text{Residual Saturation})$$

where Residual Saturation = residual oil saturation

This is a fairly rare form of organising a determination and gives rise to the complication of measuring and mapping the residual oil saturation throughout the reservoirs. It requires heavy involvement from the reservoir engineers since the measurement of residual oil saturation in waterflooding experiments on thin core plugs is largely regarded as their domain

2.7 CALCULATION OF GAS INITIALLY IN PLACE

The important parameter required in the calculation of the gas initially in place (GIIP) is the gas expansion factor, which at initial conditions is:

$$Z = \frac{PV}{nRT}$$

$$Z = 35.37 \left(\frac{P}{T} \right)$$

$$Z = 35.37 \left(\frac{P}{T} \right)$$

in which Z is the dimensionless Z-factor accounting for molecular scale effects which must be accounted for in the PVT for gas at reservoir conditions. T is the absolute temperature in degrees Rankine (= OF + 460). The gas expansion factor relates surface to reservoir volumes of gas and is related to the gas formation volume factor as:

$$V_g = \frac{V_{sc}}{Z} \left(\frac{P}{P_{sc}} \right) \left(\frac{T_{sc}}{T} \right)$$

$$V_g = \frac{V_{sc}}{Z} \left(\frac{P}{P_{sc}} \right) \left(\frac{T_{sc}}{T} \right)$$

$$V_g = \frac{V_{sc}}{Z} \left(\frac{P}{P_{sc}} \right) \left(\frac{T_{sc}}{T} \right)$$

The GIIP can then be calculated as:

$$\text{GIIP} = \frac{\text{STOIP}}{Z} (1 - \text{Residual Saturation}) \left(\frac{P}{P_{sc}} \right) \left(\frac{T_{sc}}{T} \right)$$

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3.0 MATERIAL BALANCE EQUATION

3.1 INTRODUCTION

The Schilthuis material balance equation has long been regarded as one of the basic tools of reservoir engineers for interpreting and predicting reservoir performance. In this section, the zero dimensional material balance is derived and subsequently applied, using mainly the interpretative technique of Havlena and Odeh, to gain an understanding of reservoir drive mechanisms under primary recovery conditions.

Although the classical material balance techniques, once applied, have now largely been superseded by numerical simulators, which are essentially multi-dimensional, multi-phase, dynamic material balance programs, the classical approach is well worth studying since it provides a valuable insight into the behaviour of hydrocarbon reservoirs.

3.2 DERIVATION OF THE CUMULATIVE MATERIAL BALANCE FOR OIL RESERVOIRS

The derivation of the volumetric material balance equation was first presented in 1936 by Schilthuis. To appreciate the full range of complexity in derivation, consider the case of depletion of the reservoir depicted in Fig. 3.1, which has an active aquifer and a gas cap.

Fig. 3.1. Reservoir considered in the derivation of the fully expanded material balance equation.

If N is the STOIP in the oil column and

HCPV of the gascap

$m =$

HCPV of the oil column

at initial conditions, which is therefore a constant, then applying the defining expression for the STOIP,

$$N = \frac{V_p}{B_o} = \frac{V_p}{B_o} (1 - GOR) \quad (1)$$

and

$$m = \frac{V_{gc}}{B_o} \quad (2)$$

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giving a total HCPV of $(1 + m)NBO_i$ (rb).

Suppose that at a given time after the start of production the pressure has dropped from its initial value at datum, p_i (psia), to some current average value, p (psia), as the result of fluid withdrawal from the reservoir. Then, if during this pressure drop, Δp , the entire reservoir system is allowed to expand (in an artificial sense) underground, the total expansion plus any natural water influx must equal the volume of fluids expelled from the reservoir as production. The volumetric material balance can then be expressed in reservoir barrels as:

$$NBO_i + \Delta V_{wif} = \Delta V_{p} + \Delta V_{g} + \Delta V_{o} \quad (3)$$

Perhaps the simplest way to visualise the balance is that if a measured surface volume of production (oil/gas/water) were returned to the reservoir at the reduced pressure, p , it must

fit exactly into the volume which is the sum of the total expansion and the water influx. The material balance equation can then be evaluated as follows.

3.2.1 Left-hand side (underground withdrawal - rb)

Cumulative oil production: The reservoir depicted in Fig. 3.1 is particularly complicated in that it contains a gas cap. Consequently, the initial pressure in the oil column must also be equal to the bubble point pressure, $p_i = p_b$, since the oil is gas saturated; therefore, even a slight drop in the pressure will cause the liberation of solution gas in the reservoir. The cumulative volumes of produced fluids measured at the surface are:

N_p = cumulative oil (stb)

W_p = cumulative water (stb)

G_p = cumulative gas (scf)

and:

GOR (scf/stb) = $\frac{G_p}{N_p}$

$\bar{GOR} = \frac{G_p}{N_p}$

GOR (scf/stb) = $\frac{G_p}{N_p}$

which is the cumulative or average GOR since the start of production.

Returning the produced fluids to the reservoir at the reduced pressure, p , at which the FVF's and solution GOR are: BO (rb/stb), BW (rb/stb), B_g (rb/scf) and RS (scf/stb) gives:

oil + dissolved gas = $N_p BO$ (rb)

water = $W_p BW$ (rb)

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free gas = $(G_p - N_p RS) B_g$ (rb)

The first of these contains $N_p RS$ scf of dissolved gas, consequently the free gas to be returned is $(G_p - N_p RS) B_g$ (scf). To complicate matters, some of the free gas will be that evolved from the oil column in producing below the bubble point while there may also be a component of gas cap gas which is eventually produced; the difficulty is distinguishing between the two. The cumulative underground withdrawal is therefore:

$N_p [BO + (RS - RS_p) B_g] + W_p BW$ (rb) 3.2

3.2.2 Right-hand side (expansion plus water influx)

In allowing the total system to expand as the pressure drops from p_i to p , consideration must be given to everything that can change volume during the depletion: the oil and its originally dissolved gas, the gas cap, the connate water and the reservoir pore volume; added to which is the cumulative water influx. The separate components may be accounted for as follows.

1) Oil plus originally dissolved gas: The original HCPV of the oil column is $N A h$ (rb).

Since the reservoir under study, Fig. 3.1, is initially at the bubble point, reducing the pressure will result in the release of gas and the shrinkage of oil such that its change

in volume is:

$$N(\bar{V}_g - \bar{V}_{g0}) \quad (3.3)$$

The initial volume of gas in the oil column is $\bar{V}_{g0}$ (scf) and the amount still dissolved at the reduced pressure, p , is $\bar{V}_g$ (scf). Therefore, the amount liberated is:

$$N(\bar{V}_{g0} - \bar{V}_g) \quad (3.4)$$

and the total change in volume (expansion) of the oil column is:

$$N[(\bar{V}_o - \bar{V}_{o0}) + (\bar{V}_{g0} - \bar{V}_g)] \quad (3.5)$$

2) Gas cap: The original HCPV of the gas cap is $mNBO_i$ (rb) which expressed as a surface volume is:

$$\bar{V}_{g0}$$

$$\bar{V}_g \quad (3.6)$$

$$\bar{V}_{g0}$$

and, at the reduced pressure, p , it occupies a reservoir volume of:

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$$\bar{V}_g$$

$$\bar{V}_{g0} \quad (3.7)$$

$$\bar{V}_{g0}$$

The gas cap expansion is therefore:

$$\bar{V}_{g0}$$

$$\bar{V}_{g0} (1 - \bar{V}_g / \bar{V}_{g0}) \quad (3.6)$$

$$\bar{V}_{g0}$$

3) Connate water: Although of low compressibility, the volume of connate water in the reservoir is large and therefore its expansion must be catered for during depletion. If $\bar{S}_{w0}$ is the average connate water saturation, determined as described in section 3.4, then its expansion can be calculated applying the definition of isothermal compressibility to the water as:

$$\bar{V}_{w0} = \bar{V}_w \Delta \bar{S}_w$$

where $\bar{V}_w$ is the total volume of water:

$$\bar{V}_{w0}$$

$$\bar{V}_w = \bar{V}_{w0} \bar{S}_w = \bar{V}$$

$$1 - \bar{S}_{w0} \bar{S}_w$$

The total HCPV, as defined above, is $(1 + \bar{S}_{w0}) \bar{V}_{w0}$ (rb), therefore:

$$(1 + \bar{S}_{w0}) \bar{V}_{w0} \bar{S}_w \Delta \bar{S}_w$$

$$\bar{V}_{w0} = (3.7)$$

$$1 - \bar{S}_{w0}$$

4) Pore compaction: As fluids are produced and the pressure declines the entire reservoir pore volume is reduced (compaction) and the change in volume, although

negative, expels an equal volume of fluid as production and is therefore additive to the expansion terms. Again, applying the definition of isothermal compressibility to the entire pore volume:

$$(1 + \alpha) \Delta p = \Delta V$$

$$\alpha(\Delta p) = \frac{\Delta V}{V} \Delta p = (\Delta p) \quad (3.8)$$

$$1 - \alpha \Delta p$$

5) Water influx: If the reservoir is connected to an active aquifer then, once the pressure drop is communicated either wholly or partially throughout its volume, the water will expand resulting in an influx of W_e (stb) or W_e (rb).

Adding equations 3.5, 3.6, 3.7 and 3.8, together with the influx and equating the sum to the underground withdrawal, equation 3.2, gives an expression for the Fully Expanded Material Balance as:

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$$N[(1 - \alpha) \Delta p + (\alpha - \beta) \Delta p] + W_e$$

$$= N[(1 - \alpha) \Delta p + (\alpha - \beta) \Delta p] + W_e$$

$$= N[(1 - \alpha) \Delta p + (\alpha - \beta) \Delta p] + W_e$$

$$= N[(1 - \alpha) \Delta p + (\alpha - \beta) \Delta p] + W_e$$

$$(1 + \alpha) \Delta p = \Delta V$$

$$+ W_e \quad (3.9)$$

$$1 - \alpha \Delta p$$

In this general form there are so many unknowns in the equation (section 3.4) that the attainment of an exact or unique solution is virtually impossible. Fortunately, most reservoirs are simpler than that depicted in Fig. 3.1, leading to more straightforward formulations of the equation. In terms of importance it is as powerful in reservoir engineering as Einstein's $E = mc^2$ in nuclear physics but unfortunately is a little more cumbersome in form.

If there is water or gas injection, or both, into the reservoir, the cumulative surface volumes: W_e (stb) and W_{eg} (scf) would represent equivalent reservoir volumes at pressure p of W_e (rb) and $W_{eg} \alpha_{g1}$ (rb), in which α_{g1} is the FVF of the lean injected gas and it is assumed that the water is injected free of gas. The cumulative injected volumes can be either subtracted from the left-hand side of the material balance, resulting in a net underground withdrawal, or added to the expansion terms on the right, as appropriate in the particular application.

3.3 THE MATERIAL BALANCE EXPRESSED AS A LINEAR EQUATION

Since the advent of sophisticated numerical reservoir simulation techniques, the Schilthuis material balance equation has been regarded by many engineers as being of historical interest only; a technique used back in the nineteen forties and fifties when people still used slide-rules. It is therefore interesting to note that as late as 1963-4, Havlena and Odeh presented two of the most interesting papers ever published on the subject of applying the material balance equation and interpreting the results. Their papers described the technique of interpreting the material balance as the equation of a straight line.

In an attempt to condense the material balance into a more tractable form, the nomenclature of Havlena and Odeh will be employed throughout this section. This reduces equation 3.9 to the form:

$$N_p = N_p(\text{res} + \text{res} + \text{res}) + N_p \text{res} \quad 3.10$$

in which:

$$N_p = N_p [\text{res} + (\text{res} - \text{res}) \text{res}] + N_p \text{res} (\text{res}) \quad 3.11$$

$$\text{res} = [(\text{res} - \text{res}) + (\text{res} - \text{res}) \text{res}] (\text{res}/\text{res}) \quad 3.12$$

$$\text{res}$$

$$\text{res} = \text{res} (-1) (\text{res}/\text{res}) \quad 3.13$$

$$\text{res}$$

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$$(\text{res} \text{res} + \text{res}) \Delta \text{res}$$

$$\text{res} = (1 + \text{res}) \text{res} (\text{res}/\text{res}) \quad 3.14$$

$$1 - \text{res}$$

The final three terms, having units of rb/stb, represent underground expansions per unit volume of stock tank oil.

3.4. SOLVING THE MATERIAL BALANCE (KNOWNs AND UNKNOWNs)

Considering the fully expanded material balance equation, 3.9, the parameters involved can be divided into the following categories:

Should be "known" Potential unknowns

NP N

RP We

WP p

cw BO, RS, Bg

SWC m

BW cf

Total 6 Total 8

The score of 6 knowns and 8 potential unknowns reveals the perennial difficulty not just with material balance but throughout the subject of reservoir engineering as a whole: there are far too few equations for the number of unknowns for which they must be solved. Numerical simulation does not alleviate the condition, it merely adds to the potential unknowns appearing in the sparse equations: geometry, ϕ , k, kr, etc. But rather than writing off reservoir engineering as a serious subject, the uncertainty simply adds to its allure, throwing emphasis on observation and judgement rather than numeracy for there can be rarely a unique mathematical solution to problems. Considering further the "knowns" and "unknowns."

Knowns: For strictly commercial reasons, NP (cumulative oil production) is the best "known"

of all but, frustratingly, in many older fields or fields that are in remote locations where produced gas cannot be utilised, RP (cumulative or average GOR = GP/NP) and WP (cumulative water production) are not necessarily measured. This moves them into the unknown category and exacerbates the difficulty in making sense of the material balance, if not making it an impossibility. It is assumed that the petrophysical evaluation is "always correct", in which case the average connate water saturation for the reservoir is evaluated as:

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$$\bar{S}_{wi} = \frac{\sum (h_i \phi_i S_{wi})}{\sum h_i \phi_i} \quad 3.15$$

$\bar{S}_{wi}$

which is averaged over all the wells and the subscript "in" relates to the different layers selected within each.

Unknowns: From early in the appraisal stage of a field there is always a volumetric estimate of the STOIP, N, available but as soon as there is production/pressure history for a reservoir this figure must be disregarded and an attempt made to establish it through solution of the material balance. The value so determined is the "effective" or "active" STOIP: that oil contributing to the production/pressure history, which is usually somewhat different from the volumetric estimate (usually being smaller) due to oil being trapped in undrained fault compartments or low-permeability regions of the reservoir.

3.5. COMPARISON BETWEEN MATERIAL BALANCE AND NUMERICAL SIMULATION MODELLING

It is a quite widely held belief that since the advent of numerical simulation modelling in the 1960's, material balance can now be discarded as an approximate technique of historical interest only. Nothing could be further from the truth, however, and anyone who embarks on a numerical simulation study without first applying material balance to the observed production/pressure history of a field is running the risk of error.

Fig. 3.4. Contrast between material balance and numerical simulation in history matching field performance.

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Consider an undersaturated oil reservoir in which the principle unknowns are the STOIP, N, and whether there has been a finite water influx, W_e , or not. The difference between material balance and simulation in matching field performance for such a reservoir is illustrated in Fig. 3.4.

Provided the PVT and pressure decline trend have been rationalised then they are input to material balance calculations along with the production data as known quantities, nothing else is required. Then by "adept manipulation" - as described in later sections of the chapter - an attempt is made to solve the material balance equation to determine the two principle unknowns, N and W_e , the latter representing the drive mechanism. The main advantage in this approach is that provided an average pressure decline can be defined for the system,

then no geometrical model is required. The accumulation is seen simply as an underground "blob" of hydrocarbons. In this respect the model is zero dimensional with the average pressure determined at a point (datum plane) which is representative of the reservoir as a whole. It is this unique property that permits the attempted solution of the equation to determine the hydrocarbons in place and define the drive mechanism.

The alternative approach of employing numerical simulation modelling to match production performance is also illustrated in Fig. 3.4. Now the input must contain a full geological and petrophysical description of the reservoir necessary to give physical structure to the model. But if these are defined then so too is the STOIP which is therefore input as a known quantity. Similarly a part of the physical description is the aquifer model, one is either (mathematically) attached to the reservoir or it is not, at the discretion of the engineer. Consequently, the drive mechanism is also being fed in as an input assumption or partial assumption (i.e. if the aquifer is defined through history matching with the simulator then its properties are dependent on the assumed STOIP). Therefore, since the required answer, the STOIP, must necessarily be specified from the outset, then numerical simulation applied to the history matching of field performance can hardly be regarded as an investigative technique: the results achieved being simply a reflection of the engineer's input assumptions. The remainder of the simulation input consists of the production and PVT data together with SCAL results most notably the relative permeability functions required to shift fluids from one grid block to the next. Strangely enough the pressures are not input to the model as known quantities, instead individual grid block pressures and fluid saturations comprise the "solution" achieved at the end of each time step by simultaneously solving large numbers of second order differential equations expressed in finite difference form. In this respect the simulator tackles the problem backwards: "unknowns" such as the STOIP are fed in to determine the "knowns" such as the pressure.

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